ENGINEERING
TOMORROW

Data Sheet

Liquid level sensor
Type **AKS 4100** and **AKS 4100U**

Designed specifically to measure liquid levels in a wide range of refrigeration applications

The AKS 4100/4100U liquid level sensor is designed specifically to measure liquid levels in a wide range of refrigeration applications.

The AKS 4100/4100U liquid level sensor is based on a proven techhology called Time Domain Reflectometry (TDR) or Guided Micro Wave.

AKS 4100/4100U liquid level sensor can be used to measure the liquid level of many different refrigerants in vessels, accumulators, receivers, standpipes, etc.The electrical output is a 2-wired, loop powered 4 – 20 mA output signal, which is proportional to the refrigerant liquid level.

AKS 4100/4100U in a cable version is suitable for HCFC, Non flammable HFC and R717 (Ammonia), and differing lengths from 800 mm / 31.5 in. and up to 5000 mm / 197 in.

The coaxial version of AKS 4100/4100U is designed for use with R744 (CO_2), HCFC, Non flammable HFC and R717 (Ammonia).

The AKS 4100/4100U coaxial version should always be used for marine applications for all refrigerant types. The AKS 4100/4100U cable version should NOT be used for CO_2 or marine applications.

Dust, foam, vapour, agitated surfaces, boiling surfaces, changes in density or in the dielectric constant , εr, for the liquid have no influence on the AKS 4100/4100U performance.

Oil accumulated in the bottom of a standpipe will not disturb the liquid level signal and it is not necessary to remove AKS 4100/4100U for cleaning after oil has been drained out of the standpipe.

Features

- Approved and qualified by Danfoss for refrigeration applications
- One product covering several probe lengths (cable version)
- A single product for all commonly used refrigerants (cable version)
- Cable version requires less top-end clearance for installation and service
- Proven operation with all refrigerants in combination with oil
- No need to clean cable version when fully covered by oil
- The cable version is very compact and easy to handle, ship, install and use with different lengths and refrigerants
- Changes of the liquid dielectric constant (εr) do not affect operation
- 5000 mm / 197 in. probe length with cable version
- 2-wire loop powered; no separate transformer needed
- Multi language HMI. Level and setting readout in mm,cm,m (ft, in.)
  - Language HMI versions:
    - English (default), German, French, Spanish
    - English (default),Japanese, Chinese Russian

NOTE:
AKS 4100/4100U can be connected directly to Danfoss EKE 347 liquid level controller and thus be powered from EKE 347.If used together with Danfoss EKC 347 liquid level controller, a 14 – 30 V DC supply is required.

Media

Refrigerants

The listed refrigerants are qualified and approved by Danfoss: [(1)]
R717 / NH_3: -40 / +50 °C / (-40 / +122 °F)
R744 / CO_2: -50 / +15 °C / (-58 / +59 °F)
HCFC: R22 -50 / +48 °C / (-58 / +118 °F)
HFC: R404A -50 / +15 °C / (-58 / +59 °F)
R410A -50 / +15 °C / (-58 / +59 °F)
R134A -40 / +50 °C / (-40 / +122 °F)

The listed refrigerants may be used in the complete temperature range of AKS 4100/4100U, however, the accuracy may be affected if the above listed temperature range is exceeded.

Other refrigerants within the groups of HCFC and HFC can be detected and measured if the following conditions are fulfilled:

Reference conditions
Dielectric constant
Cable version can be used in R717 / NH3, HCFC and HFC (εr, liquid > 5.6).

The coaxial version is mandatory for use in:
- R744 / CO_2 (εr, liquid > 1.3)
- Marine applications

The coaxial version can also be used in the refrigerants:
- R717 / NH_3, HCFC and HFC

New refrigerants

Danfoss products are continually evaluated for use with new refrigerants depending on market requirements.

When a refrigerant is approved for use by Danfoss, it is added to the relevant portfolio, and the R number of the refrigerant (e.g. R513A) will be added to the technical data of the code number. Therefore, products for specific refrigerants are best checked at store.danfoss.com/en/, or by contacting your local Danfoss representative.

[1] AKS 4100 Coaxial 280mm and AKS 4100U Coaxial 11 in are only released for R717/NH_3

Product specification

Pressure and temperature data

Table 1: Pressure and temperature data

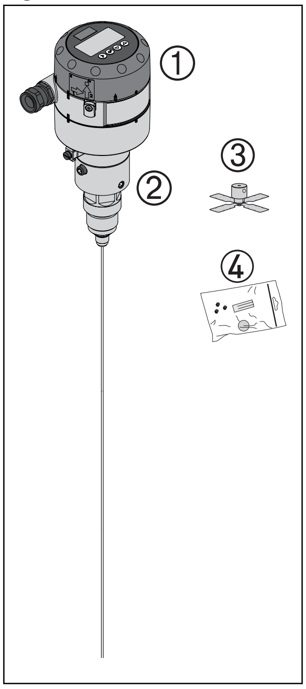
Description	Features
Refrigerant temperature	-60 °C / +100 °C (-76 °F / +212 °F)
Ambient temperature	-40 °C / +80 °C (-40 °F / +176 °F) For HMI : -20 °C / +60 °C (-4 °F / +140 °F)
Process pressure	-1 – 100 bar / -14.5 – 1450 psig
Terminals (spring loaded)	0.5 – 1.5 mm² (~20-15 AWG)
Ambient temperature supply voltage limitations:	-40 °C / +80 °C (-40 °F / +176 °F) -20 °C / +80 °C (-4 °F / +176 °F) : 14 – 30 V DC

Product concept

AKS 4100/4100U is available in two different versions:
- Cable version
- Coaxial version

Cable version

Figure 1: CABLE version

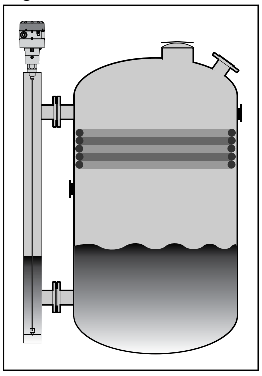


1	Signal converter, which can be supplied with or without HMI
2	Mechanical process connection with 5 m / 197 in., Ø2 mm / 0.08 in. stainless cable
3	Counterweight
4	**Accessory bag comprising:** • 3 mm set screws • Red cover to protect mechanical process connection (**2**) prior to mounting signal converter • Setting label

With the cable version it is possible to adapt the AKS 4100/4100U to any possible length in the range of 800 mm / 31.5 in. to 5000 mm / 196.9 in.

Cable version can be used in R717 / NH_3, HCFC and HFC (εr, liquid > 5.6).

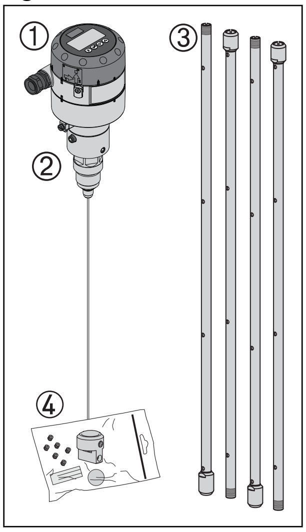
Figure 2: AKS 4100/4100U cable version must ALWAYS be installed in a standpipe



Coaxial version

Coaxial D14 version

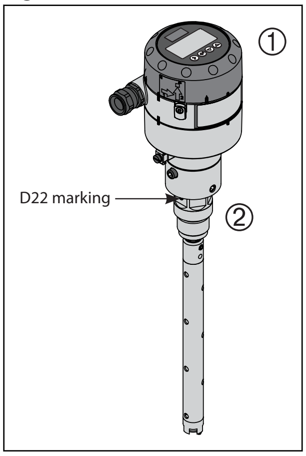
Figure 3: Coaxial D14 version



1	Signal Converter (with or without HMI)
2	Mechanical process connection with 5 m / 197 in., Ø2 mm / 0.08 stainless wire
3	Tube(s) depending on required length
4	**Accessory bag comprising:** • End Connector (incl. 3 mm / 0.12 in. set screws.) • 3 mm / 0.12 in. set crews (1 set screw pr. tube) • Red cover to protect mechanical process connection (**2**), before Signal Converter is mounted • Setting label

Coaxial D22 version

Figure 4: Coaxial D22 version



1	Signal Converter (with or without HMI)
2	Mechanical process connection 280 mm / 11 in., 8 mm / 0.3 in. inner rod

The coaxial version is mandatory for use in:

- R744 / CO_2 (εr, liquid > 1.3)
- Marine applications

The coaxial version can also be used in the refrigerants:
- R717 / NH_3, HCFC and HFC

Figure 5: AKS 4100/4100U, Coaxial can be installed in a standpipe (a) or directly in a vessel (b)

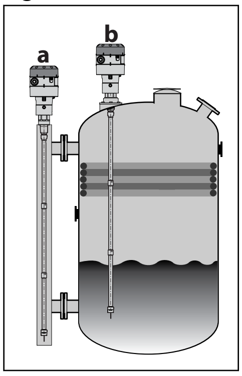


Table 2: The coaxial version is available in the following probe lengths:

Danfoss type	Tube diameter		Type selection in HMI	Thread
AKS 4100, 280 mm	22 mm	0.87 in.	D22	G1 in. pipe thread
AKS 4100, 500 mm	14 mm	0. 55 in.	D14	G1 in. pipe thread
AKS 4100, 800 mm	14 mm	0.55 in.	D14	G1 in. pipe thread
AKS 4100, 1000 mm	14 mm	0.55 in.	D14	G1 in. pipe thread
AKS 4100, 1200 mm	14 mm	0.55 in.	D14	G1 in. pipe thread
AKS 4100, 1500 mm	14 mm	0.55 in.	D14	G1 in. pipe thread
AKS 4100, 1700 mm	14 mm	0.55 in.	D14	G1 in. pipe thread
AKS 4100, 2200 mm	14 mm	0.55 in.	D14	G1 in. pipe thread
AKS 4100U, 11.0 in.	22 mm	0.87 in.	D22	¾ in. NPT
AKS 4100U, 19.2 in.	14 mm	0.55 in.	D14	¾ in. NPT
AKS 4100U, 30 in.	14 mm	0.55 in.	D14	¾ in. NPT
AKS 4100U, 45 in.	14 mm	0.55 in.	D14	¾ in. NPT
AKS 4100U, 55 in.	14 mm	0.55 in.	D14	¾ in. NPT
AKS 4100U, 65 in.	14 mm	0.55 in.	D14	¾ in. NPT
AKS 4100U, 85 in.	14 mm	0.55 in.	D14	¾ in. NPT

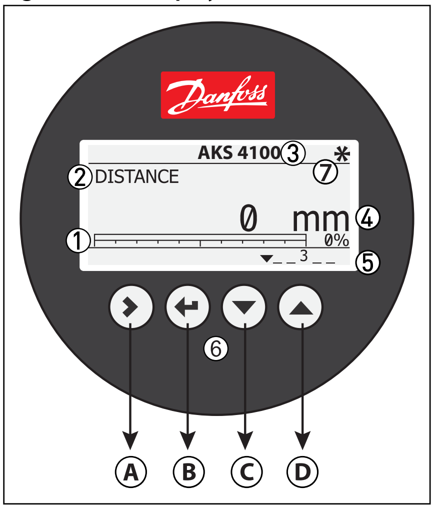
Optional HMI

The optional HMI Service/Display unit is used for commissioning and quick on-site setup and is easily mounted on the AKS 4100/4100U.

The service unit supports mulitple languages in both SI and US units.

Supported standard languages: English (default), German, French, Spanish, Japanese, Chinese and Russian.

Figure 6: HMI display unit



1	4 – 20 mA output displayed as bar graph and in percentage [%]
2	Measurement name (in this example, DISTANCE)
3	Device tag name
4	Measurement reading and unit
5	Device status (markers) **Marker 1, 2 and 3 (Error)** Hardware problem; the Signal Converter hardware is defective. Contact Danfoss **Marker 4 and 5 (Notification)** Depending on the level, the marker is ON or OFF. Used for Danfoss service information only
6	Keypad buttons
7	Flashing star indicating unit in operation
A	Enter menu system Enter QUICK SETUP
B	Unit change at distance/level readout: m, cm, mm, in, ft
C & D	Change between: Distance (1) Level (2) Output (%) (3) Output (mA) (4)

(1) If the display is set to “DISTANCE” the displayed value will be the distance from the Reference point to the top surface of the liquid refrigerant (see Page 11 and Page 12)
(2) If the display is set to “LEVEL” then the value displayed will be: PROBE LENGTH (entered in QUICK SETUP) – DISTANCE (see Page 11 and Page 12)
(3) Will represent the level of refrigerant,in percent, scaled (entered in QUICK SETUP) according to: SCALE 4 mA (0%), SCALE 20 mA (100%) (see Page 11 and Page 12).
(4) Will represent the level of refrigerant, in 4 – 20 milliampere, scaled (entered in QUICK SETUP) according to: SCALE 4 mA (4 mA), SCALE 20 mA (20 mA) (see Page 11 and Page 12).

Design

Table 3: Design

Description	Probe types	Values
Options	Cable	Mechanical process connection with 5 m / 197 in., Ø2 mm / 0.08 in. stainless cable: Mechanical thread on the mechanical process connection AKS 4100: G1 in. pipe thread. Aluminium gasket included AKS 4100U: ¾ in. NPT
	Coaxial D14	Mechanical process connection with 5 m / 197 in., Ø2 mm / 0.08 in. stainless cable and 14 mm / 0.55 in. outer stainless tube: Mechanical thread on the mechanical process connection AKS 4100: G1 in. pipe thread. Aluminium gasket included AKS 4100U: ¾ in. NPT Stainless steel tubes supporting the available probe length
	Coaxial D22	Mechanical process connection with in 22 mm / 0.87 in. outer stainless tube. 8 mm / 0.3 in. inner rod. Mechanical thread on the mechanical process connection AKS 4100: G1 in. pipe thread. Aluminium gasket included AKS 4100U: ¾ in. NPT
	LCD display	
Insertions (probe) length	Coaxial D14	AKS 4100: 500, 800, 1000, 1200, 1500, 1700 and 2200 mm AKS 4100U: 19.2, 30, 45, 55, 65, 85 in.
	Coaxial D22	AKS 4100: 280 mm AKS 4100U: 11.0 in.
	Single cable Ø2 mm / 0.08 in.: 800 – 5000 mm / 31.5-197 in.	
Dead zone	This depends on the type of probe. (see pages 7 and 8)	

Table 4: Display and User interface

Description	Values
Display	Integrated LCD display 128 × 64 pixels in 8-step greyscale with 4-button keypad
Interface languages	English (default), German, French, Spanish, Japanese, Chinese, Russian

Table 5: Operating conditions

Description	Values
Ambient temperature	-40 °C / +80 °C (-40 °F / +175 °F) For HMI: -20 °C / +60 °C (-4 °F / +140 °F)
Storage temperature	-40 °C / +85 °C (-40 °F / +185 °F)
Process connection temperature	Standard: -60 °C / +100 °C (-76 °F / +212 °F)
Operating pressure	Standard: -1 bar to 100 bar / -14.5 psig to 1450 psig

Table 6: Other conditions:

Description	Values
Liquid dielectric constant (εr)	Cable version to be used in R717 / NH3, HCFC and HFC εr, liquid > 5.6 Coaxial version is mandatory in R744 / CO2 εr, liquid > 1.3
Vibration resistance	EN 60721-3-4 (1 – 9 Hz: 3 mm / 10 – 200 Hz:1g; 10g shock half-wave sinusoidal: 11 ms)
Protection category	IP 66/67 equivalent to NEMA type 4X (housing) and type 6P (probe)

Table 7: Installation conditions

Description	Values
Dimensions and weights	See Dimensions and weights

Table 8: Material

Description	Values
Housing	Aluminium
Coaxial D14 and D22 version	Standard: Stainless steel (1.4404 / 316L)
Single cable	Standard: Stainless steel (1.4401 / 316)
Process fitting	Standard: Stainless steel (1.4404 / 316L)
Gaskets	EPDM (-50 / +150 °C (-58 / +300 °F)
Cable gland	Plastic (black)

Table 9: Process connections - Thread

Description	Values
Single cable Ø2 mm / 0.08¨	AKS 4100: G1 inch pipe thread. Aluminium gasket included AKS 4100U: ¾ in. NPT
Coaxial D14 and D22 version	AKS 4100: G1 inch pipe thread. Aluminium gasket included AKS 4100U: ¾ in. NP

Table 10: Electrical connections

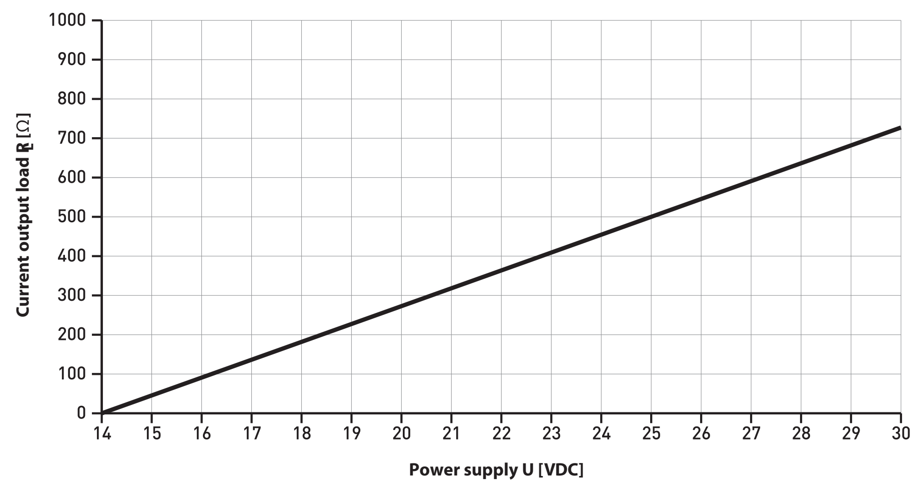
Description	Values
Power supply	Terminals output: 14 – 30 V DC. Min./Max. Value for an output of 22 mA at the terminal. Ambient temperature limitations: -40 °C / +80 °C (-40 °F / +176 °F) : 16 – 30 V DC -20 °C / +80 °C (-4 °F / +176 °F) : 14 – 30 V DC
Current output load	RL [Ω] ≤ ((Uext -14 V)/20 mA). – Default (Error output set to 3.6 mA) RL [Ω] ≤ ((Uext -14 V)/22 mA). – (Error output set to 22 mA)
Cable gland	AKS 4100: PG 13, M20×1.5 ; (cable diameter: 6 – 8 mm / 0.24 – 0.31 in.) AKS 4100U: ½ in. NPT
Cable entry capacity (terminal)	0.5 – 1.5 mm^2 (~20-15 AWG)

Table 11: Input and output

Description	Values
Output signal	4 – 20 mA or 3.8 – 20.5 mA acc. to NAMUR NE 43
Resolution	±3 µA
Temperature drift	Typically 75 ppm/K
Error signal	High: 22 mA; Low: 3.6 mA acc. to NAMUR NE 43; Hold (frozen value - not available with NAMURNE 43 compliant output

Minimum power supply voltage

Figure 7: Use this graph to find the minimum power supply voltage for a given current output load:



NOTE:

Minimum power supply voltage for an output of 22mA at the terminal.

Measuring system

Table 12: Measuring system

Features	Description
Measuring principle	2-wire loop-powered level transmitter; Time Domain Reflectometry (TDR)
Application range	Level measurement of liquid refrigerants. Approved refrigerants: Halogen Free / Environmently friendly: R717 / NH3, R744 / CO_2 HCFC and non flammable HFC.
Primary measured value	Time between the emitted and received signal
Secondary measured value	Distance or level

Measuring principle (Cable and Coaxial)

The AKS 4100/4100U electronic converter emits low-intensity, high frequency electromagnetic pulses with a width of approximately 1 nanosecond, which travel at the speed of light along the probe (wire or coaxial cable) down to the liquid surface.

The pulses are reflected by the liquid surface, guided back along the probe, and received and analysed by the AKS 4100/4100U electronic converter and then converted into a liquid level reading. This method is called time domain reflectometry (TDR) or guided microwave.

The dielectric constant, εr, of the liquid is a key parameter and has a direct impact on the degree of reflection of the high frequency electromagnetic pulses. Liquids with high εr values, such as ammonia, produce strong reflections, while liquids with low εr values, such as CO_2, produce weak reflections.

As long as the εr value of the liquid refrigerant is higher than 1.2, AKS 4100/4100U can detect the liquid level and level measurement accuracy is not affected.

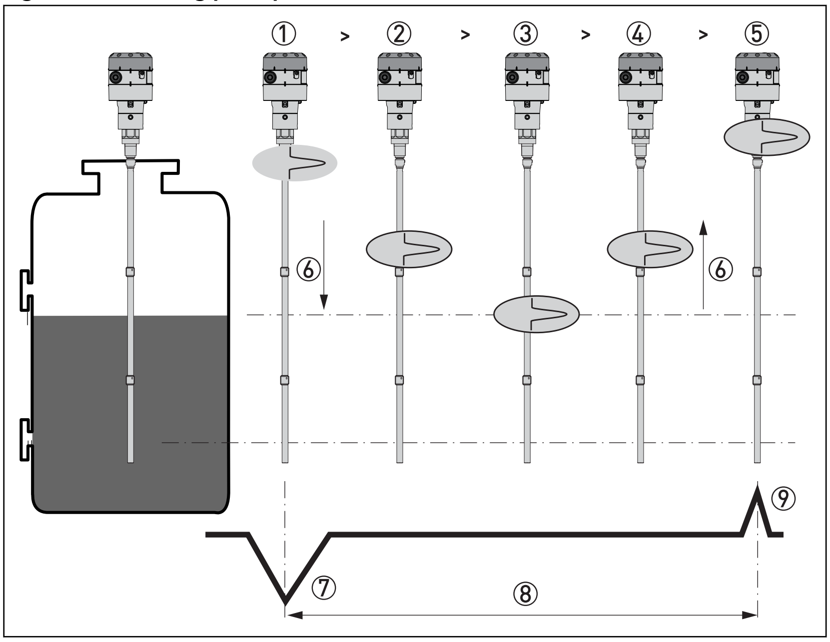
NOTE:

If the temperature condition in the standpipe /vessel is known, a constant (dielectric constant of the refrigerant gas) can be entered (parameter 2.5.3 GAS EPS.R), in order to obtain improved Top and Bottom Dead Zone values.

Refer to Page 11 page 10 and Page 12 for Measuring range of AKS 4100/4100U - CABLE version and COAXIAL version.

For details of gas constant values for different temperatures and refrigerants plus the procedure for entering these via the HMI, refer to Page 23 and Page 24.

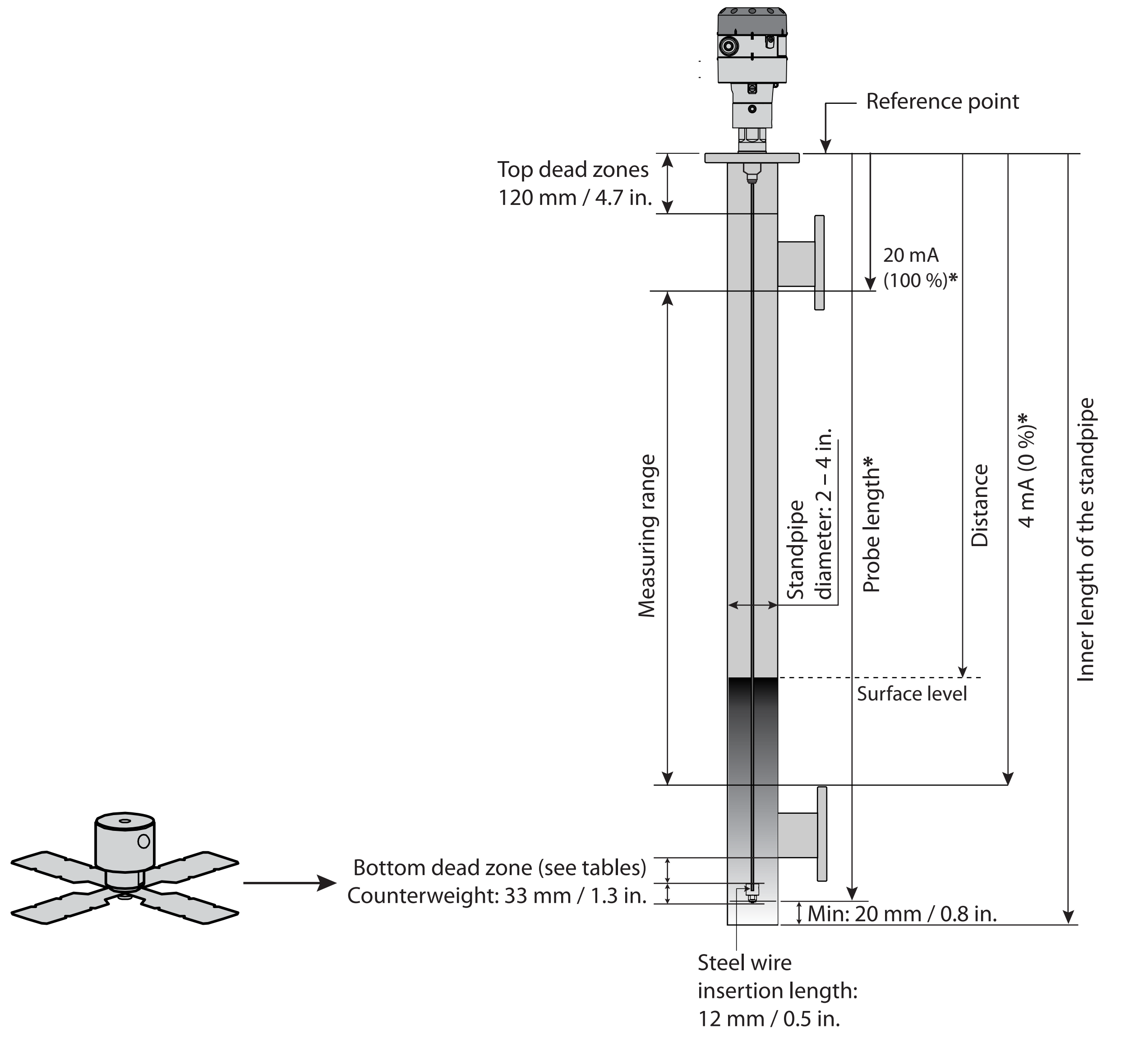
Figure 8: Measuring principle (Cable and Coaxial)



1	The electromagnetic (EM) pulse is transmitted by the signal converter
2	The pulse goes down the probe at the speed of light in air, V1
3	The pulse is reflected
4	The pulse goes up the probe at speed, V1
5	The converter receives the pulse and records the signal
6	The EM pulse moves at speed, V1
7	Transmitted EM pulse
8	Half of this time is equivalent to the distance from the reference point of the device (the flange facing) to the surface of the product
9	Received EM pulse

Measuring range of AKS 4100/4100U - CABLE version

Figure 9: Measuring range of AKS 4100/4100U - CABLE version



NOTE:
* Values to be entered into HMI Quick Setup menu and recorded on the setting label. Stick the setting label onto the Signal Converter either inside or outside.

Table 13: Bottom deadzone values based on the factory setting of dielectric constant

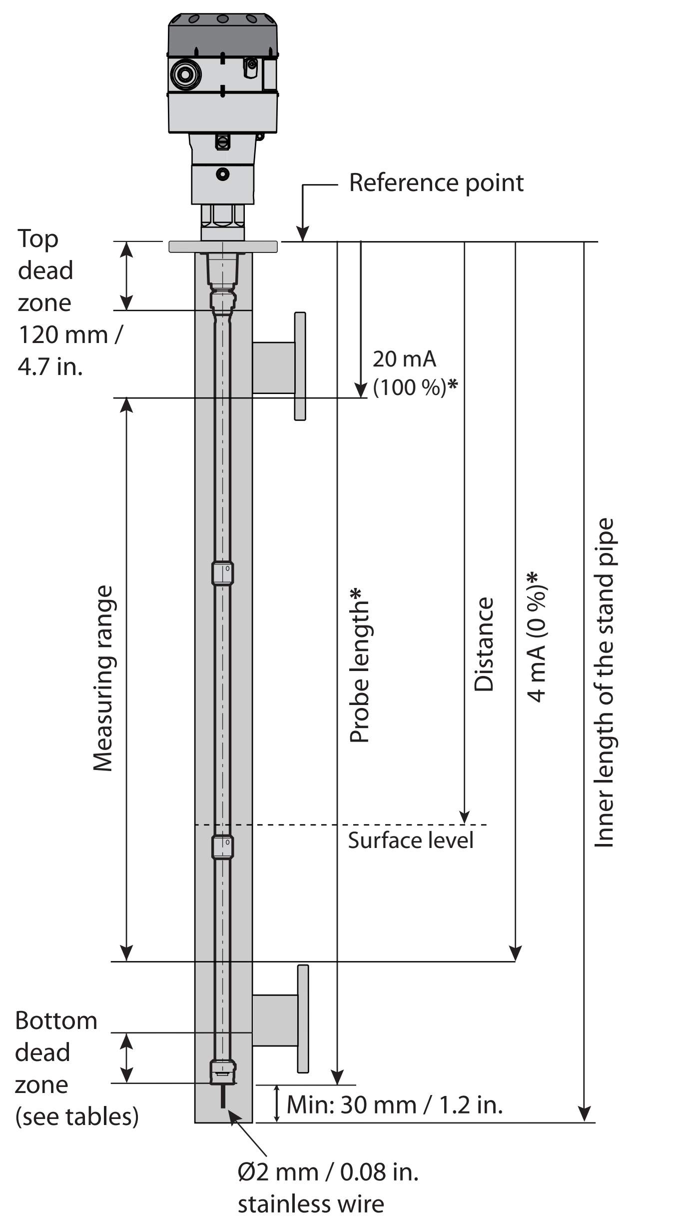
Refrigerant	Probe length range		Bottom dead zone	
	[mm]	[in.]	[mm]	[in.]
Ammonia, HFC, HCFC	800	31.5	115	4.2
	801 – 999	31.5 – 39	120	4.7
	1000 – 1999	39 – 79	150	5.9
	2000 – 2999	79 – 118	180	7.1
	3000 – 3999	118 – 157	210	8.3
	4000 – 5000	157 – 197	240	9.4

Table 14: Improved Bottom dead zone values after the adjustment of dielectric constant

Refrigerant	Probe length range		Bottom dead zone	
	[mm]	[in.]	[mm]	[in.]
Ammonia, HFC, HCFC	800 – 5000	31.5 – 197	90	3.5

Measuring range of AKS 4100/4100U - COAXIAL D14 version

Figure 10: Measuring range of AKS 4100/4100U - COAXIAL D14 version



NOTE:
* Values to be entered into HMI Quick Setup menu and recorded on the setting label. Stick the setting label onto the Signal Converter either inside or outside.

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Table 15: Dielectric Constant εr always set during Quick Setup

Refrigerant	Probe length		Bottom Dead Zone	Bottom Dead Zone
	[mm]	[in.]	[mm]	[in.]
CO_2	500	19.7	170	6.7
	800	31.5		
	1000	39.4		
	1200	47.2		
	1500	59.1		
	1700	66.9		
	2200	86.6		

Table 16: Factory setting

Refrigerant	Probe length		Bottom Dead Zone	Bottom Dead Zone
	[mm]	[in.]	[mm]	[in.]
Ammonia	500	19.7	95	3.7
	800	31.5	104	4.1
	1000	39.4	110	4.3
	1200	47.2	116	4.6
	1500	59.1	125	4.9
	1700	66.9	131	5.2
	2200	86.6	146	5.8

Table 17: Improved Bottom dead zone values after the adjustment of dielectric constant

Refrigerant	Probe length		Bottom Dead Zone	Bottom Dead Zone
	[mm]	[in.]	[mm]	[in.]
Ammonia	500	19.7	80	3.2
	800	31.5		
	1000	39.4		
	1200	47.2		
	1500	59.1		
	1700	66.9		
	2200	86.6		

Table 18: Factory setting

Refrigerant	Probe length		Bottom Dead Zone	Bottom Dead Zone
	[mm]	[in.]	[mm]	[in.]
HCFC,HFC	500	19.7	115	4.5
	800	31.5	124	4.9
	1000	39.4	130	5.1
	1200	47.2	136	5.4
	1500	59.1	145	5.7
	1700	66.9	151	5.9
	2200	86.6	166	6.5

Table 19: Improved Bottom dead zone values after the adjustment of dielectric constant

Refrigerant	Probe length		Bottom Dead Zone	Bottom Dead Zone
	[mm]	[in.]	[mm]	[in.]
HCFC,HFC	500	19.7	100	3.9
	800	31.5		
	1000	39.4		
	1200	47.2		
	1500	59.1		
	1700	66.9		
	2200	86.6		

NOTE:

It is mandatory to input dielectric constant for CO_2 applications.

AKS 4100U

Table 20: Dielectric Constant εr always set during Quick Setup

Refrigerant	Probe Length	Bottom Dead Zone	Bottom Dead Zone
	[in.]	[in.]	[mm]
CO_2	19.2	6.7	170
	30		
	45		
	55		
	65		
	85		

Table 21: Factory setting

Refrigerant	Probe Length [in.]	Bottom Dead Zone [in.]	Bottom Dead Zone [mm]
Ammonia	19.2	3.73	95
	30	4.05	103
	45	4.5	114
	55	4.8	122
	65	5.1	130
	85	5.7	145

Table 22: Improved Bottom dead zone values after the adjustment of dielectric constant

Refrigerant	Probe Length [in.]	Bottom Dead Zone [in.]	Bottom Dead Zone [mm]
Ammonia	19.2	3.1	80
	30		
	45		
	55		
	65		
	85		

Table 23: Factory setting

Refrigerant	Probe Length [in.]	Bottom Dead Zone [in.]	Bottom Dead Zone [mm]
HCFC,HFC	19.2	4.52	115
	30	4.84	123
	45	5.29	134
	55	5.59	142
	65	5.89	150
	85	6.49	165

Table 24: Improved Bottom dead zone values after the adjustment of dielectric constant

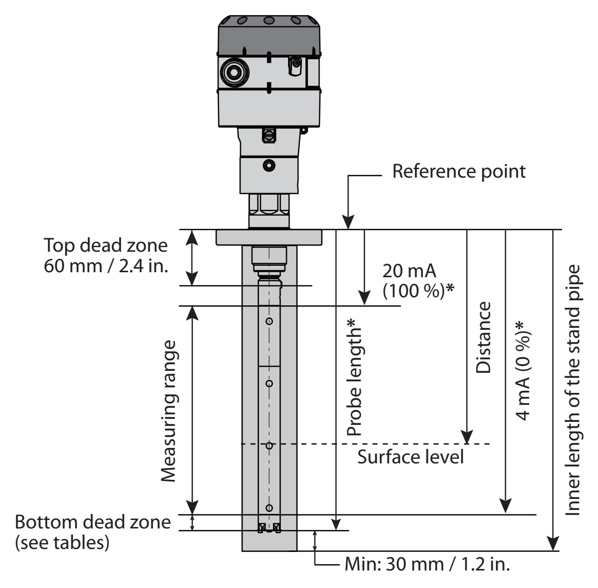
Refrigerant	Probe Length [in.]	Bottom Dead Zone [in.]	Bottom Dead Zone [mm]
HCFC,HFC	19.2	3.94	100
	30		
	45		
	55		
	65		
	85		

NOTE:

It is mandatory to input dielectric constant for CO_2 applications.

Measuring range of AKS 4100/4100U - COAXIAL D22 version

Figure 11: Measuring range of AKS 4100/4100U - COAXIAL D22 version



NOTE:
* Values to be entered into HMI Quick Setup menu and recorded on the setting label. Stick the setting label onto the Signal Converter either inside or outside.

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Table 25: Factory setting

Refrigerant	Probe length		Bottom Dead Zone	Bottom Dead Zone
	[mm]	[in.]	[mm]	[in.]
Ammonia	280	11	48	1.9

Table 26: Improved Bottom dead zone values after the adjustment of dielectric constant

Refrigerant	Probe length		Bottom Dead Zone	Bottom Dead Zone
	[mm]	[in.]	[mm]	[in.]
Ammonia	280	11	40	1.6

AKS 4100U

Table 27: Factory setting

Refrigerant	Probe Length	Bottom Dead Zone	Bottom Dead Zone
	[in.]	[in.]	[mm]
Ammonia	11	1.9	48

Table 28: Improved Bottom dead zone values after the adjustment of dielectric constant

Refrigerant	Probe Length	Bottom Dead Zone	Bottom Dead Zone
	[in.]	[in.]	[mm]
Ammonia	11	1.6	40

Connections

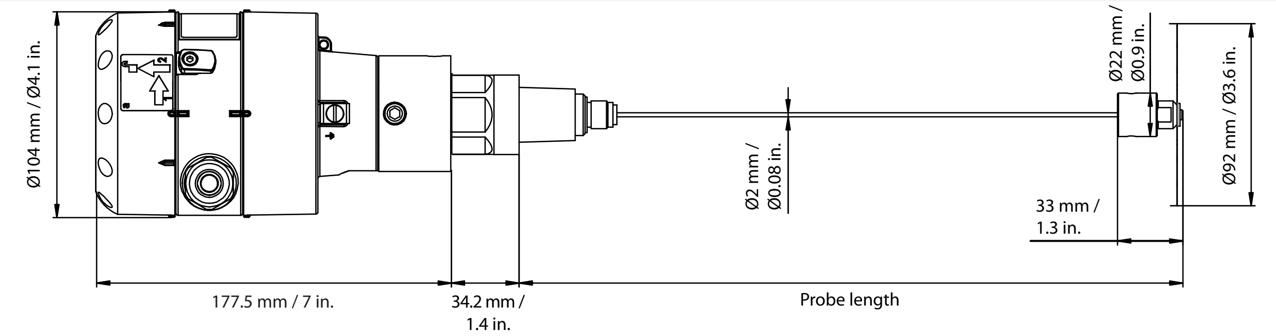
Mechanical connection
Cable version / Coaxial version:

- AKS 4100
  - G1 in. pipe thread
  - Aluminium gasket included
- AKS 4100U
  - ¾ in. NPT

Dimensions and weights

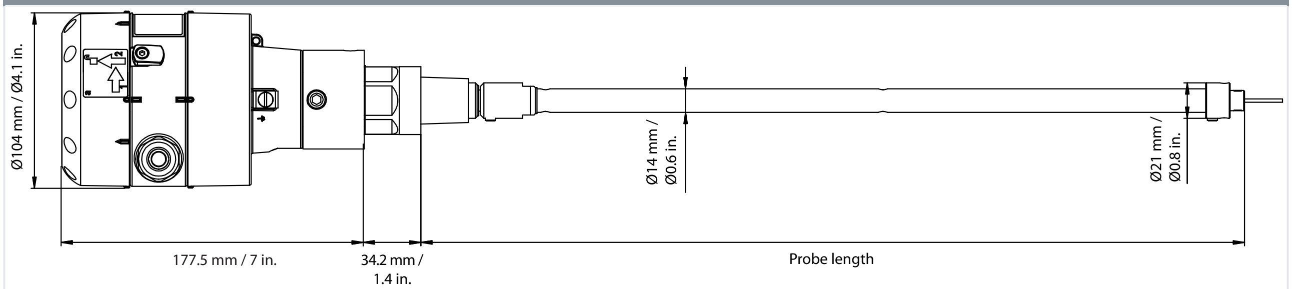
CABLE version

Table 29: CABLE version

Weight: approx. 2.3 kg / 5.1 lbs


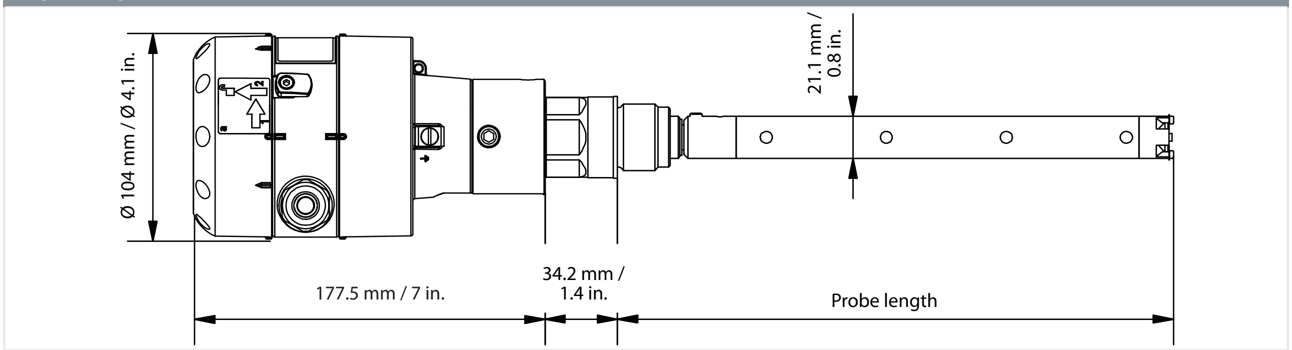
COAXIAL D14 version

Table 30: Coaxial D14 version

Weight: approx. 3.8 kg / 8.4 lbs


COAXIAL D22 version

Table 31: Coaxial D22 version

Weight: 2.4 kg / 5.3 lbs


Ordering

Cable version

Figure 12: Cable version

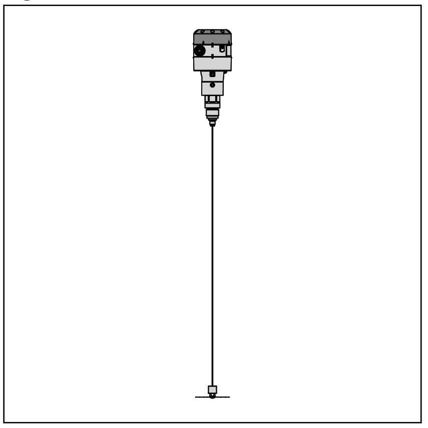


Table 32: Cable version

Description	Refrigerant	Code number with HMI English (default) German French Spanish	Code number with HMI English (default) Japanese Chinese Russian	Code number Without HMI*
AKS 4100 with 5 m / 197 in., Ø2 mm / Ø0.08 in. stainless cable and counterweight	Ammonia, R134A, R404A, R410A, R22	084H4501	084H4550	084H4500
AKS 4100U with 5 m / 197 in., Ø2 mm / Ø0.08 in. stainless cable and counterweight	Ammonia, R134A, R404A, R410A, R22	084H4521	084H4571	084H4520

Coaxial version (available in predefined lengths, with or without HMI)

Figure 13: Coaxial version D14

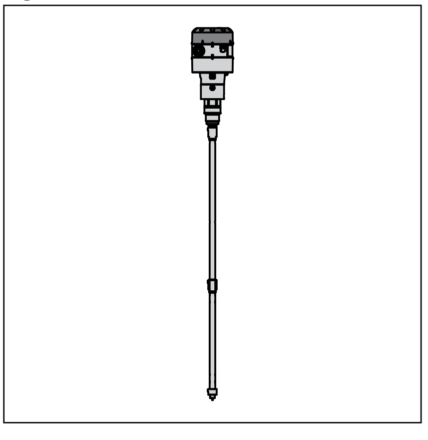


Figure 14: Coaxial version D22

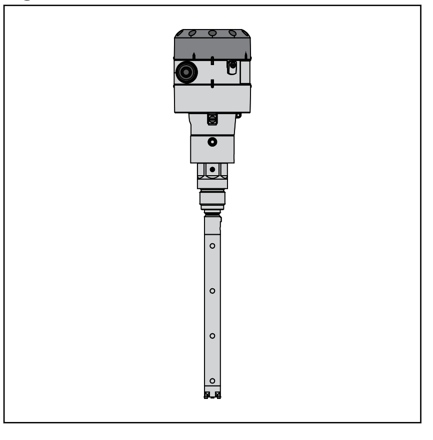


Table 33: Coaxial version - D14

Description	Probe length		Refrigerant	Code number with HMI English (default) German French Spanish	Code number with HMI English (default) Japanese Chinese Russian	Code number Without HMI*
	mm	in.				
AKS 4100 - Coaxial D14	500		Ammonia, CO_2, R134A, R404A, R410A, R22	084H4510	084H4560	084H4503
AKS 4100 - Coaxial D14	800		Ammonia, CO_2, R134A, R404A, R410A, R22	084H4511	084H4561	084H4504
AKS 4100 - Coaxial D14	1000		Ammonia, CO_2, R134A, R404A, R410A, R22	084H4512	084H4562	084H4505
AKS 4100 - Coaxial D14	1200		Ammonia, CO_2, R134A, R404A, R410A, R22	084H4513	084H4563	084H4506
AKS 4100 - Coaxial D14	1500		Ammonia, CO_2, R134A, R404A, R410A, R22	084H4514	084H4564	084H4507
AKS 4100 - Coaxial D14	1700		Ammonia, CO_2, R134A, R404A, R410A, R22	084H4515	084H4565	084H4508
AKS 4100 - Coaxial D14	2200		Ammonia, CO_2, R134A, R404A, R410A, R22	084H4516	084H4566	084H4509

Description	Probe length		Refrigerant	Code number with HMI English (default) German French Spanish	Code number with HMI English (default) Japanese Chinese Russian	Code number Without HMI*
	mm	in.				
AKS 4100U - Coaxial D14		19.2	Ammonia, CO_2, R134A, R404A, R410A, R22	084H4530	084H4580	084H4524
AKS 4100U - Coaxial D14		30	Ammonia, CO_2, R134A, R404A, R410A, R22	084H4531	084H4581	084H4525
AKS 4100U - Coaxial D14		45	Ammonia, CO_2, R134A, R404A, R410A, R22	084H4532	084H4582	084H4526
AKS 4100U - Coaxial D14		55	Ammonia, CO_2, R134A, R404A, R410A, R22	084H4533	084H4583	084H4527
AKS 4100U - Coaxial D14		65	Ammonia, CO_2, R134A, R404A, R410A, R22	084H4534	084H4584	084H4528
AKS 4100U - Coaxial D14		85	Ammonia, CO_2, R134A, R404A, R410A, R22	084H4535	084H4585	084H4529

Table 34: Coaxial version - D22

Description	Probe length		Refrigerant	Code number with HMI English (default) German French Spanish	Code number with HMI English (default) Japanese Chinese Russian	Code number Without HMI*
	mm	in.				
AKS 4100 - Coaxial D22	280		Ammonia	084H4517	084H4567	084H4518
AKS 4100U - Coaxial D22		11	Ammonia	084H4536	084H4586	084H4537
AKS 4100 - Coaxial D22	280		CO_2, R134A, R404A, R410A, R22	084H4572	084H4573	084H4574
AKS 4100U - Coaxial D22		11	CO_2, R134A, R404A, R410A, R22	084H4575	084H4576	084H4577

HMI display unit

When ordering without HMI please observe:
NOTE: **Each AKS 4100/AKS 4100 must always be programmed via the HMI display unit.**

The HMI display unit can be ordered separately:

- **084H4540 / 084H4590** AKS 4100/4100U HMI display unit with rear cover and mounting bracket. The mounting bracket is very useful when the AKS 4100/4100U have to be programmed. The same AKS 4100/4100U HMI display unit can be used to programme more AKS 4100/4100U and both Cable and Coaxial versions.
- **084H4548 / 084H4598** AKS 4100/4100U HMI display unit (usually spare part).

Accessories

Table 35: Accessories

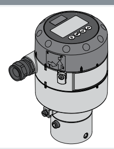
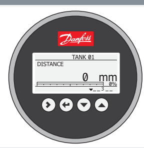
AKS 4100/4100U excluding cable gland	HMI Service/Display unit	
a) 	b) 	c) 

Table 36: Accessories

Description	Code number with HMI English (default) German French Spanish	Code number with HMI English (default) Japanese Chinese Russian
AKS 4100/4100U HMI Service/Display unit with rear cover and mounting bracket - **c)**	084H4540	084H4590
AKS 4100/4100U HMI Display - **b)**	084H4548	084H4598
AKS 4100/4100U Signal Converter + Metaglass with HMI, excluding cable gland - **a)**	084H4555	084H4556
AKS 4100/4100U converter connecting cable (5 pcs.)	084H4557	

Service kits

Table 37: Service kits

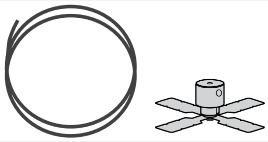
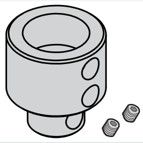
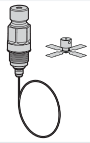
Cable and counterweight - CABLE version	End connector incl screws - COAXIAL D14 version	Process connection, counterweight and cable
a) 	**b)** 	**c)** 

Table 38: Service kits

Description	Content	Code number
Cable and counterweight for AKS 4100/4100U - CABLE version - **a)**	Cable - 5 m / 197 in., Ø2 mm / Ø0.08 in.	084H4542
	Crimp	
	Counterweight	
End connector incl screws for AKS 4100/4100U - COAXIAL D14 version - **b)**	End connector (incl. 3 mm / 0.12 in. set screws)	084H4549
Process connection, counterweight and 5 m / 197 in., Ø2 mm / Ø0.08 in. cable for AKS 4100 - CABLE and COAXIAL D14 version - **c)**	1 in. process connection	084H4545
	Counterweight	
Process connection, counterweight and 5 m / 197 in., Ø2 mm / Ø0.08 in. cable for AKS 4100U - CABLE and COAXIAL D14 version - **c)**	¾ in. NPT process connection	084H4546
	Counterweight	

Other spare parts

Table 39: Other spare parts

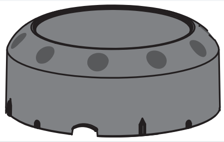
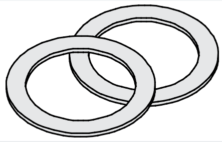
Other spare parts			
a)	**b)** 	**c)** 	**d)** 

Table 40: Other spare parts

Description	Code number
AKS 4100/4100U Coaxial tube. Tube length: 680 mm / 26.8 in. - **a)**	084H4543
AKS 4100/4100U blank top cover for signal converter - **b)**	084H4544
AKS 4100/4100U Aluminium gaskets (10 pcs.) for 1 in. process connection - **c)**	084H4547
AKS 4100 1 in. welding connection	027F1010
Process connection AKS 4100 - Coaxial D22 - G1 in. - 280 mm. For Ammonia only - **d)**	084H4551
Process connection AKS 4100U - Coaxial D22 - ¾ in. NPT – 11 in. For Ammonia only - **d)**	084H4552
Process connection AKS 4100 - Coaxial D22 - G1 in. - 280 mm. For CO_2, R134A, R404A, R410A, R22 only - **d)**	084H4558
Process connection AKS 4100 - Coaxial D22 - ¾ in. NPT – 11 in. For CO_2, R134A, R404A, R410A, R22 only - **d)**	084H4559

Quick setup

Table 41: Quick Setup

Quick Setup (all values below are only examples)

- Connect the device to the power supply (see the section "Electrical installation/ connection").
- Press › 3 times .

```
AKS 4100
QUICK SETUP ?
YES          NO
```

- Press ›

```
AKS 4100
PROBE TYPE
SINGLE CABLE
```

- Press ▼ or ▲ to select between SINGLE, COAXIAL D14 and COAXIAL D22.
  Choose **SINGLE** and press ↵ to confirm.

```
AKS 4100
PROBE  LENGTH
05000 mm
```

- Press › to change the PROBE LENGTH.
  Press › to change the position of the cursor.
  Press ▼ to decrease the value or ▲ to increase the value.
  Press ↵ to confirm.

```
AKS 4100
SCALE 4 mA
04946 mm
```

- Press › to change of SCALE 4 mA.
  Press › to change the cursor position.
  Press ▼ to decrease the value or ▲ to increase the value.
  Press ↵ to confirm.

```
AKS 4100
SCALE 20 mA
00070 mm
```

- Press › to change of SCALE 20 mA.
  Press › to change the cursor position.
  Press ▼ to decrease the value or ▲ to increase the value.
  Press ↵ to confirm.

```
AKS 4100
QUICK SETUP
COMPLETED IN  8
```

- Wait for QUICK SETUP to complete
  8-second timeout

```
AKS 4100
1.0.0
QUICK SETUP
```

- Press ↵ to confirm.

```
AKS 4100
1.0.0
STORE NO
```

- Press ▼ or ▲ to select either STORE NO or STORE YES.
  Press ↵ to confirm.

Default screen appears:

```
AKS 4100
DISTANCE
5000 mm
```

Quick Setup completed

You have the possibility of checking your settings by pressing › twice.

```
AKS 4100
SINGLE CABLE          5000   mm
(0%) 4 mA             4877 mm
(100%) 20 mA          120 mm
```

Press ↵ ▲ ↵ to return to default screen.

NOTE:

The signal converter can be programmed with or without mechanical process connector assembled.

When CO_2 is used:

Table 42: Quick Setup (all values below are only examples)

Quick Setup (all values below are only examples)

- Connect the device to the power supply (see the section "Electrical installation/ connection").
- Press (›) 3 times .

```
AKS 4100
QUICK SETUP ?
YES            NO
```

- Press (›)

```
AKS 4100
PROBE TYPE
SINGLE CABLE
```

- Press (▼) or (▲) to select between SINGLE, COAXIAL D14 and COAXIAL D22. Choose **COAXIAL D14** and press (↵) to confirm.

```
AKS 4100
LIQUID CO2 ?
YES            NO
```

- Press (›) (YES) to confirm

```
AKS 4100
GAS EPS R ?
001.000
```

- Press (›) to change GAS EPS.R. (Select the correct value from the tables on page 8)
  Press (›) to change cursor-position.
  Press (▼) to decrease the value or (▲) to increase the value.
- Press (↵) to confirm.

```
AKS 4100
PROBE  LENGTH
05000 mm
```

- Press (›) to change the PROBE LENGTH.
  Press (›) to change the position of the cursor.
  Press (▼) to decrease the value or (▲) to increase the value.
  Press (↵) to confirm.

```
AKS 4100
SCALE 4 mA
04946 mm
```

- Press (›) to change of SCALE 4 mA.
  Press (›) to change the cursor position.
  Press (▼) to decrease the value or (▲) to increase the value.
  Press (↵) to confirm.

```
AKS 4100
SCALE 20 mA
00070 mm
```

- Press (›) to change of SCALE 20 mA.
  Press (›) to change the cursor position.
  Press (▼) to decrease the value or (▲) to increase the value.
  Press (↵) to confirm.

```
AKS 4100
QUICK SETUP
COMPLETED IN  8
```

- Wait for QUICK SETUP to complete.
  Count down from 8 sec.

```
AKS 4100
1.0.0
QUICK SETUP
```

- Press (↵) to confirm.

```
AKS 4100
1.0.0
STORE NO
```

- Press (▼) or (▲) to select between STORE NO or STORE YES.
  Press (↵) to confirm.

  Default screen appears:

```
AKS 4100
DISTANCE
5000 mm
```

Quick Setup completed

You have the possibility of checking your settings by pressing (›) twice.

```
AKS 4100
COAXIAL D14          2200 mm
(0 %) 4 mA           1900 mm
(100 %) 20 mA          70 mm
```

Press (↵) (▲) (↵) to return to default screen.

NOTE:
The signal converter can be programmed with or without mechanical process connector assembled.

For all other refrigerants

Table 43: For all other refrigerants

For all other refrigerants		
• Connect the device to the power supply (see the section "Electrical installation/ connection"). • Press ▶ 3 times . AKS 4100 QUICK SETUP ? YES NO • Press ▶ AKS 4100 PROBE TYPE SINGLE CABLE • Press ▼ or ▲ to select between SINGLE, COAXIAL D14 and COAXIAL D22. Choose **COAXIAL D14** and press ↵ to confirm. AKS 4100 LIQUID CO2 ? YES NO • Press ▲ (NO) to confirm AKS 4100 PROBE LENGTH 05000 mm	• Press ▶ to change the PROBE LENGTH. Press ▶ to change the position of the cursor. Press ▼ to decrease the value or ▲ to increase the value. Press ↵ to confirm. AKS 4100 SCALE 4 mA 04946 mm • Press ▶ to change of SCALE 4 mA. Press ▶ to change the cursor position. Press ▼ to decrease the value or ▲ to increase the value. Press ↵ to confirm. AKS 4100 SCALE 20 mA 00070 mm • Press ▶ to change of SCALE 20 mA. Press ▶ to change the cursor position. Press ▼ to decrease the value or ▲ to increase the value. Press ↵ to confirm.	AKS 4100 QUICK SETUP COMPLETED IN 8 • Wait for QUICK SETUP to complete. Count down from 8 sec. AKS 4100 1.0.0 QUICK SETUP • Press ↵ to confirm. AKS 4100 1.0.0 STORE NO • Press ▼ or ▲ to select between STORE NO or STORE YES. Press ↵ to confirm. Default screen appears: AKS 4100 DISTANCE 5000 mm **Quick Setup completed**

NOTE:
Please note that Coaxial D22 version can only be used in R717/NH_3)

CABLE and COAXIAL version

Table 44: CABLE and COAXIAL version

Forcing mA output (all values below are only examples)		
Default screen AKS 4100 / DISTANCE / 5000 mm • Press ▶ AKS 4100 / 1.0.0 / QUICK SETUP • Press ▲ AKS 4100 / 2.0.0 / SUPERVISOR • Press ▶ AKS 4100 / 2.0.0 / ________ Enter password: ▶ ↵ ▼ ▲ ▶ ↵ AKS 4100 / 2.1.0 / INFORMATION	• Press ▲ AKS 4100 / 2.2.0 / TESTS • Press ▶ AKS 4100 / 2.2.1 / SET OUTPUT • Press ▶ AKS 4100 / SET OUTPUT / 3.5 mA • Press ▼ to decrease the value or ▲ to increase the value. Press ↵ to confirm. AKS 4100 / SET OUTPUT / 8 mA	• Press ↵ 4 times to return to default screen. Default screen appears: AKS 4100 / DISTANCE / 5000 mm **Force mA completed and disabled**

Entering refrigerant dielectric gas constant

Optional Procedure

Optional Procedure If the temperature condition in the stand pipe is known, a constant (dielectric constant of the refrigerant gas) can be entered (parameter 2.5.3 GAS EPS.R), in order to obtain lower Top and Bottom Dead Zone values (see Measuring range of AKS 4100/4100U - CABLE version and Measuring range of AKS 4100/4100U - COAXIAL D14 version).

Table 45: Entering refrigerant dielectric gas constant

Entering refrigerant dielectric gas constant (all values below are only examples)

Default screen

```
AKS 4100
DISTANCE
5000 mm
```

- Press ▶

```
AKS 4100
1.0.0
QUICK SETUP
```

- Press ▲

```
AKS 4100
2.0.0
SUPERVISOR
```

- Press ▶

```
AKS 4100
2.0.0
________
```

Enter password:
▶ ↵ ▼ ▲ ▶ ↵

```
AKS 4100
2.1.0
INFORMATION
```

- Press ▲ 4 times .

```
AKS 4100
2.5.0
APLICATION
```

- Press ▶

```
AKS 4100
2.5.1
TRACING VEL.
```

- Press ▲ 2 times .

```
AKS 4100
2.5.3
GAS EPS. R
```

- Press ▶ to change GAS EPS.R.
  (Select the correct value from the tables on page 16)
  Press ▶ to change cursor-position.
  Press ▼ to decrease the value or ▲ to increase the value.

```
AKS 4100
GAS EPS. R
1.066
```

- Press ↵ to confirm.

```
AKS 4100
2.5.3
GAS EPS. R
```

- Press ↵ 3 times .

```
AKS 4100
1.0.0
STORE NO
```

- Press ▼ or ▲ to select between STORE NO or STORE YES.
  Select STORE YES by pressing ↵

Default screen appears:

```
AKS 4100
DISTANCE
5000 mm
```

Entering the dielectric constant of refrigerant gas completed

Saturated vapour dielectric constant (default value: 1.066)

Temperature range: -60 / +50 °C (-76 / +122 °F)

Table 46: R717 (NH_3)

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1
-41 – -18	42 – 0	1.01
-17 – -5	1 – 23	1.02
-4 – 4	24 – 39	1.03
5 – 12	40 – 54	1.04
13 – 18	55 – 64	1.05
19 – 24	65 – 75	1.06
25 – 28	76 – 82	1.07
29 – 33	83 – 91	1.08
34 – 37	92 – 99	1.09
38 – 40	100 – 104	1.1
41 – 44	105 – 111	1.11
45 – 47	112 – 117	1.12
48 – 50	118 – 122	1.13

Temperature range: -60 / +48 °C (-76 / +118 °F)

Table 47: R22

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -50	-76 – -58	1
-49 – -25	57 – -13	1.01
-24 – -10	-12 – 14	1.02
-9 – 0	15 – 32	1.03
1 – 8	33 – 46	1.04
9 – 15	47 – 59	1.05

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
16 – 21	60 – 70	1.06
22 – 26	71 – 79	1.07
27 – 31	80 – 88	1.08
32 – 35	89 – 95	1.09
36 – 39	96 – 102	1.1
40 – 42	103 – 108	1.11
43 – 45	109 – 113	1.12
46 – 48	114 – 118	1.13

Temperature range: -65 / +15 °C (-85 / +59 °F)

Table 48: R410A

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-65 – -47	-85 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -13	-1 – 9	1.05
-12 – -8	10 – 18	1.06
-7 – -4	19 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 4	33 – 40	1.09
5 – 7	41 – 45	1.1
8 – 10	46 – 50	1.11
11 – 12	51 – 54	1.12
13 – 15	55 – 59	1.13

Temperature range: -60 / +15 °C (-76 / +59 °F)

Table 49: R507

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -48	-76 – -54	1.01
-47 – -36	-53 – -32	1.02
-35 – -28	-31 – -18	1.03
-27 – -21	-17 – -6	1.04
-20 – -15	-17 – -5	1.05
-14 – -10	-4 – 14	1.06
-9 – -6	13 – 22	1.07
-5 – -2	23 – 29	1.08
-1 – 2	30 – 36	1.09
3 – 5	37 – 41	1.1
6 – 8	42 – 47	1.11
9 – 11	48 – 52	1.12
12 – 13	53 – 56	1.13
14 – 15	57 – 59	1.14

Temperature range: -56 / +15 °C (-69 / +59 °F)

Table 50: R744 (CO_2)

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-56.0 – -42.0	-69 – -43	1.01
-41.0 – -28.0	-42 – -18	1.02
-27.0 – -17.0	-17 – 2	1.03
-16.0 – -9.0	3 – 16	1.04
-8.0 – -3.0	17 – 27	1.05
-2.0 – 2	28 – 36	1.06
3 – 7	37 – 45	1.07

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
8 – 11	46 – 52	1.08
12 – 14	53 – 58	1.09
15	59	1.1

Temperature range: -60 / +50 °C (-76 / + 122 °F)

Table 51: R134a

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1
-41 – -18	-42 – -0	1.01
-17 – -4	1 – 25	1.02
-3 – 5	26 – 41	1.03
6 – 13	42 – 56	1.04
14 – 20	57 – 68	1.05
21 – 25	69 – 77	1.06
26 – 30	78 – 86	1.07
31 – 34	87 – 94	1.08
35 – 38	95 – 100	1.09
39 – 42	101 – 108	1.1
43 – 45	109 – 113	1.11
46 – 48	114 – 119	1.12
49 – 50	120 – 122	1.13

Temperature range: -60 / +15 °C (-76 / +59 °F)

Table 52: R404A

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -47	-76 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -14	-1 – 7	1.05
-13 – -9	8 – 16	1.06
-8 – -4	17 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 3	33 – 38	1.09
4 – 6	39 – 43	1.1
7 – 9	44 – 49	1.11
10 – 12	50 – 54	1.12
13 – 15	55 – 59	1.13

How to change the language setting (Default: English)

Table 53: How to change the language setting

How to change the language setting (Default: English)

Default screen

AKS 4100
DISTANCE
5000 mm

- Press ►

AKS 4100
1.0.0
QUICK SETUP

- Press ▲

AKS 4100
2.0.0
SUPERVISOR

- Press ►

AKS 4100
2.0.0
____

Enter password:
► ↵ ▼ ▲ ► ↵

AKS 4100
2.1.0
INFORMATION

- Press ▲ 6 times

AKS 4100
2.7.0
DISPLAY

- Press ►

AKS 4100
2.7.1
LANGUAGE

- Press ►

AKS 4100
LANGUAGE
ENGLISH

- Press ▼ or ▲ to see the language possibilities
  Press ↵ to confirm.

AKS 4100
2.7.1
LANGUAGE

- Press ↵ 3 times

AKS 4100
2.0.0
STORE NO

- Press ▼ or ▲ to select between STORE NO or STORE YES.
  Select STORE YES by pressing ↵
  Default screen appears:

AKS 4100
DISTANCE
5000 mm

Language setup completed

Reset to factory setting

- Go to SUPERVISOR menu (see CABLE and COAXIAL version)
- Go to parameter 2.9.4 Reset Factory
- Select RESET FACTORY YES
- Press ↵ 3 times to return to default screen
- **Factory reset completed**

Certificates, declarations, and approvals

The list contains all certificates, declarations, and approvals for this product type. Individual code number may have some or all of these approvals, and certain local approvals may not appear on the list.

Some approvals may change over time. You can check the most current status at danfoss.com or contact your local Danfoss representative if you have any questions.

Table 54: Valid approvals

File name	Document type	Document topic	Approval authority
GOST FR.C.29.004.A 51938	Measuring - Performance Certificate		
UA.1O146.D.00075-19	UA Declaration	EMCD/LVD	LLC CDC EURO-TYSK
033F0689.AA	EU Declaration	EMC	Danfoss
MD 033F0686.AH	Manufacturers Declaration	PED	Danfoss
033F0695.AA	Manufacturers Declaration	China RoHS	Danfoss
0F18749.513467890YTN	Pressure - Safety Certificate	CRN	TSSA
0F19272.2	Pressure - Safety Certificate	CRN	TSSA

Table 55: Approvals and certification

CE	This device fufills the statutory requirements of the EMC directives. The manufacturer certifies successful testing of the product by applying the CE mark.
	Valid for AKS 4100 - **Not valid for AKS 4100U:** Pattern Approval Certificate of Measuring Instruments for the Russian Federation
EAC	Valid for AKS 4100 - **Not valid for AKS 4100U:** In compliance with EMC regulations in the Russian Federation
EMC	EMC Directives 2004 / 108 / EC and 93 / 68 / EEC in conjunction with EN 61326-1 (2006) and EN 61326-2-3 (2006). The device conforms to these standards if: - the device has a coaxial probe or - the device has a single probe that is installed in a metallic tank
LVD	Low-Voltage Directives 2006 / 95 / EC and 93 / 68 / EEC in conjunction with EN 61010-1 (2001)
NAMUR	NAMUR NE 21 Electromagnetic Compatibility (EMC) of Industrial Process and Laboratory Control Equipment
	NAMUR NE 43 Standardization of the Signal Level for the Failure Information of Digital Transmitters

ENGINEERING TOMORROW

Online support

Danfoss offers a wide range of support along with our products, including digital product information, software, mobile apps, and expert guidance. See the possibilities below.

The Danfoss Product Store

The Danfoss Product Store is your one-stop shop for everything product related—no matter where you are in the world or what area of the cooling industry you work in. Get quick access to essential information like product specs, code numbers, technical documentation, certifications, accessories, and more.
Start browsing at store.danfoss.com.

Find technical documentation

Find the technical documentation you need to get your project up and running. Get direct access to our official collection of data sheets, certificates and declarations, manuals and guides, 3D models and drawings, case stories, brochures, and much more.

Start searching now at www.danfoss.com/en/service-and-support/documentation.

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Find your local Danfoss website here: www.danfoss.com/en/choose-region.

Spare Parts

Get access to the Danfoss spare parts and service kit catalog right from your smartphone. The app contains a wide range of components for air conditioning and refrigeration applications, such as valves, strainers, pressure switches, and sensors.

Download the Spare Parts app for free at www.danfoss.com/en/service-and-support/downloads.

Coolselector®2 - find the best components for you HVAC/R system

Coolselector®2 makes it easy for engineers, consultants, and designers to find and order the best components for refrigeration and air conditioning systems. Run calculations based on your operating conditions and then choose the best setup for your system design.

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ENGINEERING
TOMORROW

Installation Guide

en **AKS 4100 / AKS 4100U Liquid Level Sensor - Cable version**

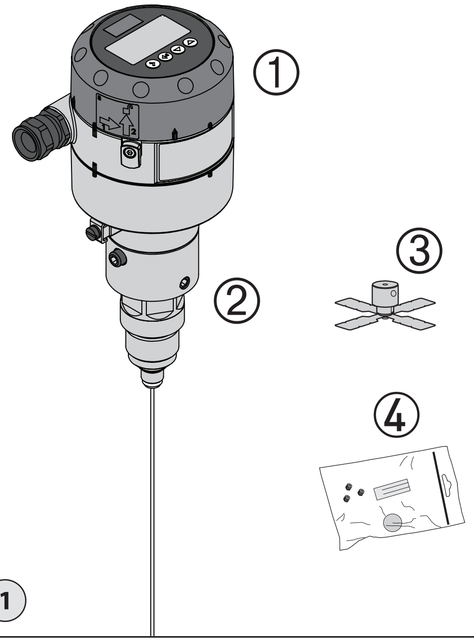
zh **液位传感器 AKS 4100 / AKS 4100U - 电缆版本**

jp **タイプ AKS 4100 / 4100U ケーブル バージョン**

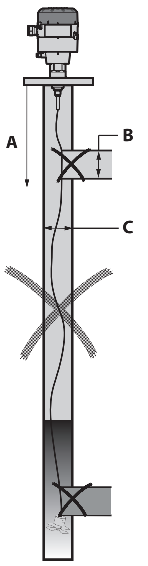
ru **Уровнимер AKS 4100 Тросовая версия**

089 0879

148R9621



1

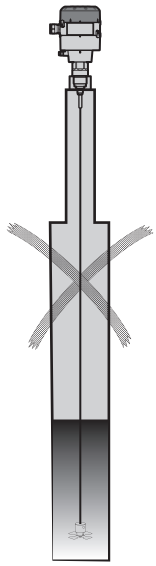


A:
Detection delay
检测延迟
ディテクションディレー
Задержки Обнаружения

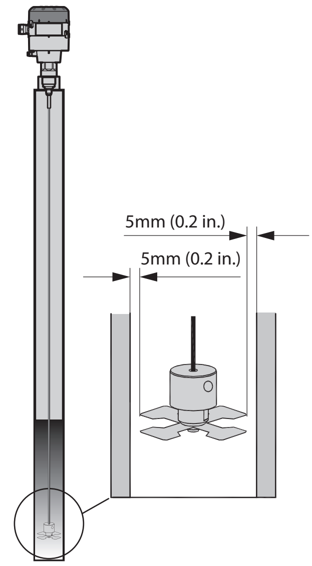
B:
Side connection pipe diameter
旁通管直径
検知の遅延
Диаметр присоединительного патрубка

C:
Stand pipe diameter
立管直径
直立管径
Диаметр измерительной колонки

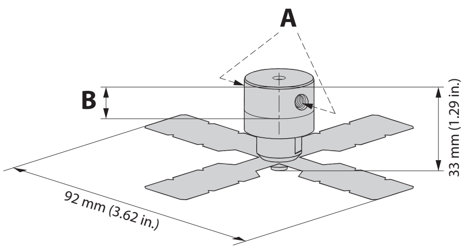
2a



2b

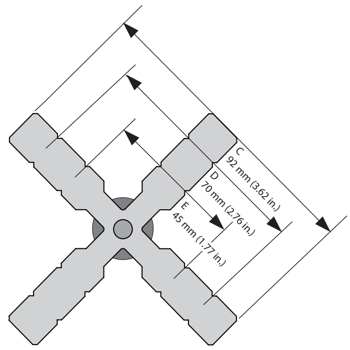


2c



A:
2 screws - 3 mm Allen key.
Torgue : 5-7 Nm (3.7-5.2 lb/ft)
使用2 颗螺钉及3 mm 内六角
扭矩： 5-7 Nm (3.7-5.2 lb/ft)
取付 ネジ 2 本 - 3 mm 六角レンチ
トルク:5～7 Nm (3.7～5.2 lb/ft)
2 винта - ключ на 3 мм.
Усилие : 5-7 Нм (3.7-5.2 lb/ft)

B:
Steel wire insertion length:
12 mm / 0.47 in.
钢缆插入长度
12 mm / 0.47 in.
鋼線 挿入長
12 mm / 0.47 in.
Длина для закрепления троса
12 mm / 0,47 in.



Standpipe/Columm 立管/竖管 **直立管 / 柱 Измерительная колонка**

C		D		E	
2 in.		3 in.		4 in.	
Pipe / 管道 / インチ管 / Труба		Pipe / 管道 / インチ管 / Труба		Pipe / 管道 / インチ管 / Труба	

Inner diameter / 内径 / 内径 / Внутренний диаметр :

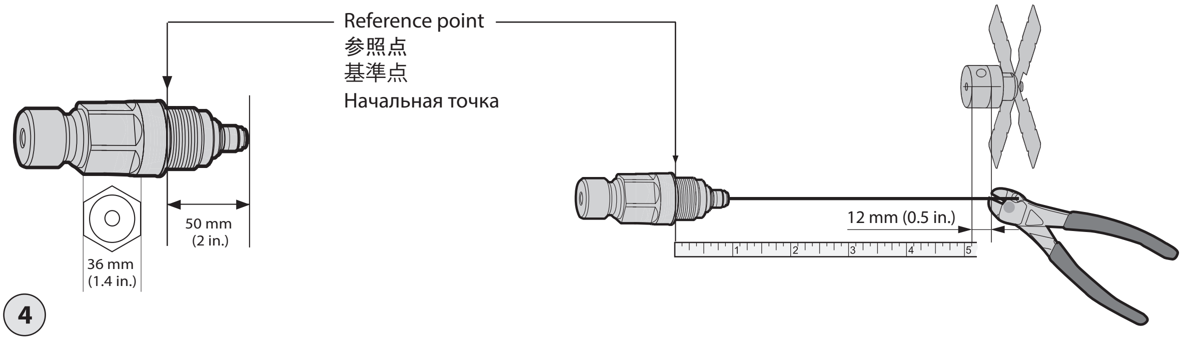
[mm]	[in.]	[mm]	[in.]	[mm]	[in.]
52	2.05	78	3.07	102*	4.02*

Total length of guided blades after cutting / 切断后导叶的总长度 /
切断後のガイド付き羽根の全長 / Длина лопастей груза после обрезки :

[mm]	[in.]	[mm]	[in.]	[mm]	[in.]
45	1.77	70	2.76	92*	3.62*

* No need to cut guided blades / 无需切断导叶/ ガイド付きブレードは切断不要 /
Нет необходимости в обрезке лопастей груза

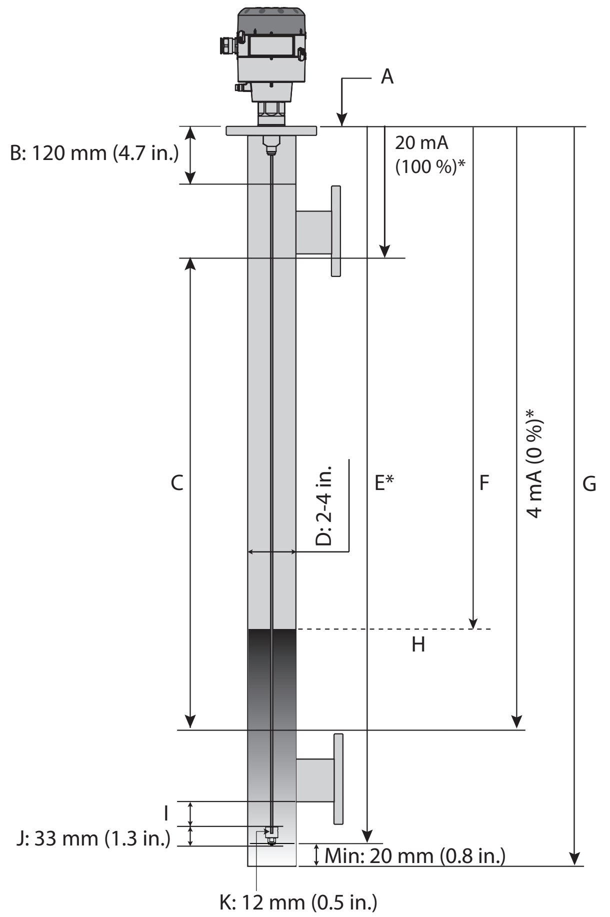
3



4

Імпортер:ТОВ з ії "Данфосс ТОВ" 04080, Київ 80, п/с 168, Україна

Info for UK customers only : Danfoss Ltd., 22 Wycombe End, HP9 1NB, GB



5

	en	zh	jp	ru
A	Reference point	参照点	基準点	Точка отсчета
B	Top dead zone	顶端死区	上部デッドゾーン	Верхние мертвые зоны
C	Measuring range	测量范围	計測範囲	Диапазон измерения
D	Stand pipe diameter	立管直径	直立管径	Диаметр стояка
E	Probe length	探头长度	プローブ長	Длина датчика
F	Distance	距离	距離	Расстояние
G	Inner length of stand-pipe	立管内部长度	直立管の内部長さ	Внутренняя длина стояка
H	Surface level	液位	液面レベル	Поверхностный уровень
I	Buttom dead zone (see tables)	底端死区（参见表格）	下部デッドゾーン (表を参照	Нижняя мертвая зона (см. таблицы)
J	Counter weight	平衡重	釣り合いおもり	Противовес
K	Steel wire insertion length	钢线插入长度	鋼線挿入長	Длина прокладки стального провода

* Values to be entered into HMI Quick Setup menu and recorded on the setting label.
Stick the setting label onto the Signal Converter either inside or outside.

将数值输入到 HMI 快速设置菜单，并在设置标签上标明。
将设置标签粘贴到信号转换器上，内侧或外侧均可。

HMI クイックセットアップメニュー及び設定ラベルに記録された値で登録されます。
信号変換器の内部または外部のどちらかに設定ラベルを貼り付けます。

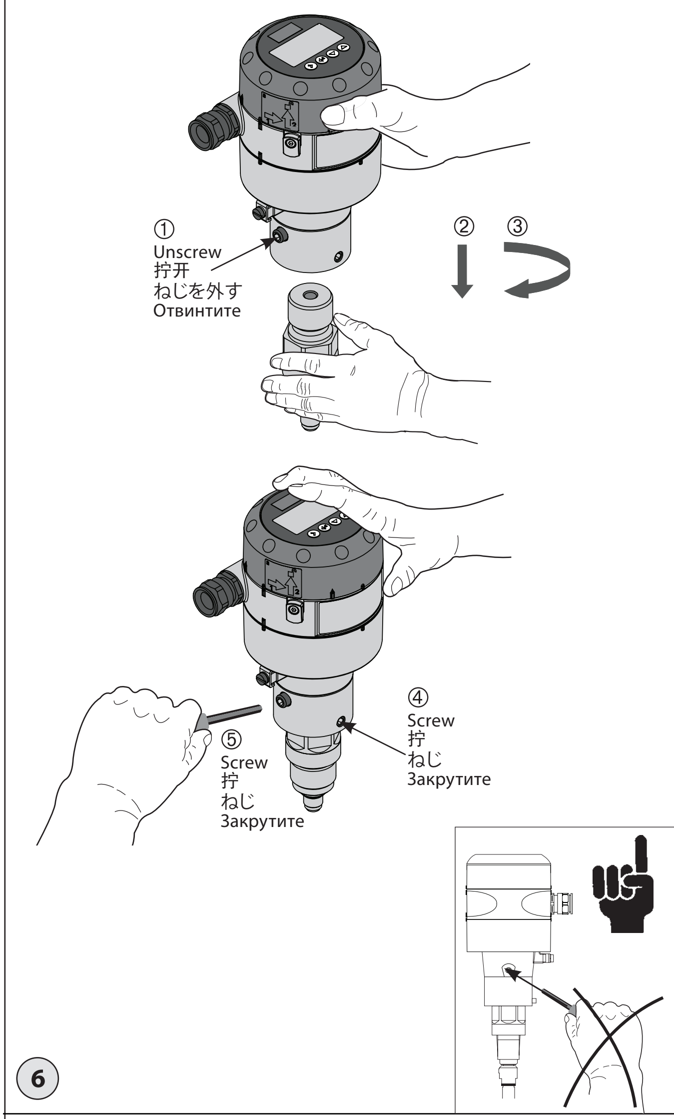
Значения должны быть введены в меню «Быстрая настройка» интерфейса «человек-машина» и записаны на бирке уровнемера. Закрепите бирку на преобразователе сигналов.

Bottom deadzone values based on the factory setting of dielectric constant
底端死区值取决于出厂时设定的介电常数值
下部デッドゾーンの値は工場出荷時 の誘電率設定に基づきます
Заводская настройка уровнимера AKS 4100 тросовая версия

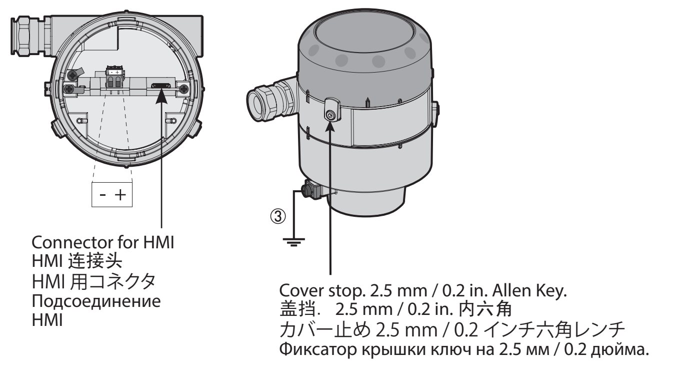
Refrigerant 制冷剂 **冷媒** **Хладагент**	**Probe length range** 探头长度范围 **プローブ長範囲** **Длина троса**		**Bottom dead zone** 底端死区 **下部デッドゾーン** **Нижняя мертвая зона**	
	[mm]	[in.]	[mm]	[in.]
Ammonia 氨 アンモニア Аммиак HFC, HCFC	800	31.5	115	4.2
	801 – 999	31.5 – 39	120	4.7
	1000 – 1999	39 – 79	150	5.9
	2000 – 2999	79 – 118	180	7.1
	3000 – 3999	118 – 157	210	8.3
	4000 – 5000	157 – 197	240	9.4

Improved Bottom dead zone values after the adjustment of dielectric constant
调整介电常数后，底端死区值有所改善
誘電率調整後の改良された下部デッドゾーンの値
Оптимизированые значения мертвой зоны после введения значения диэлектрической проницаемости

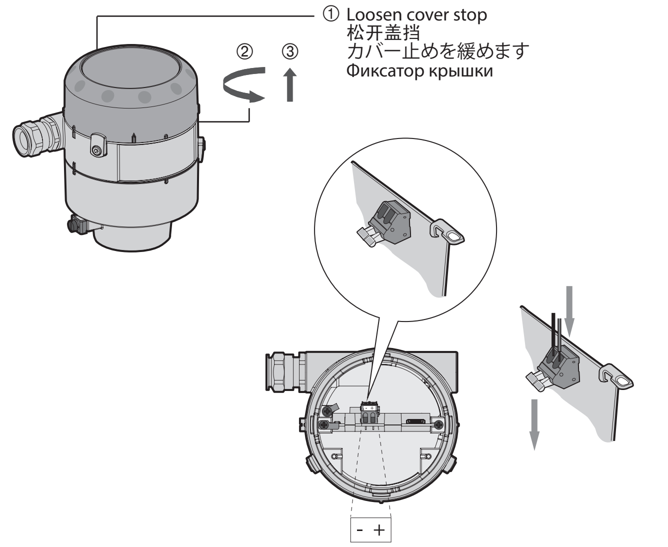
Ammonia 氨 アンモニア Аммиак HFC, HCFC	800 – 5000	31.5 – 197	90	3.5



6



7



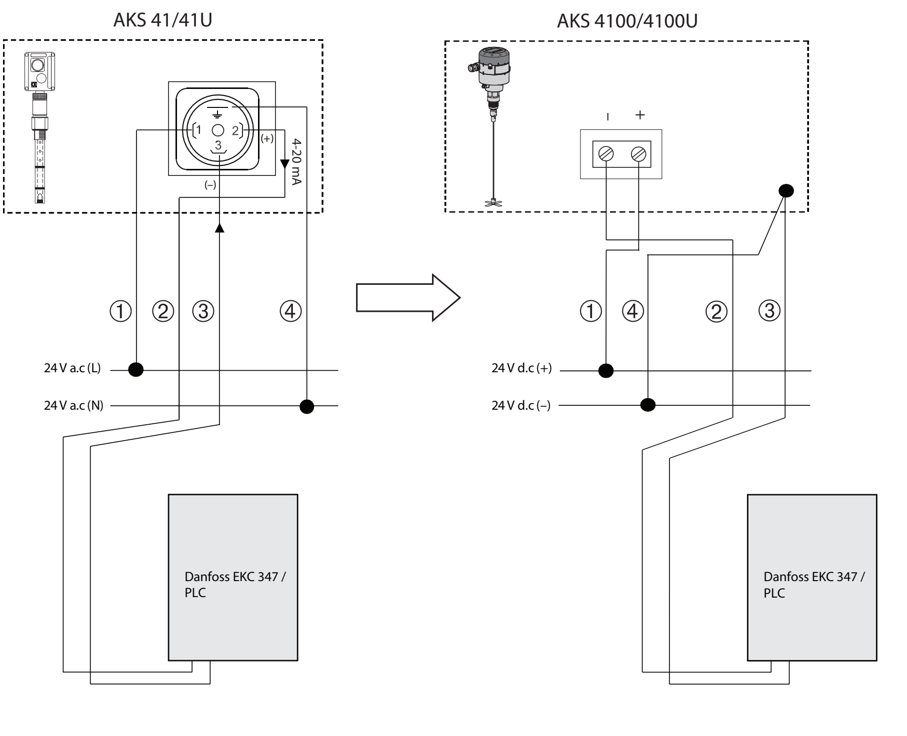
8

AKS 41/41U with a.c supply to
AKS 4100/4100U with d.c supply.

AKS 41/41U（交流电源）至
AKS 4100/4100U（直流电源）

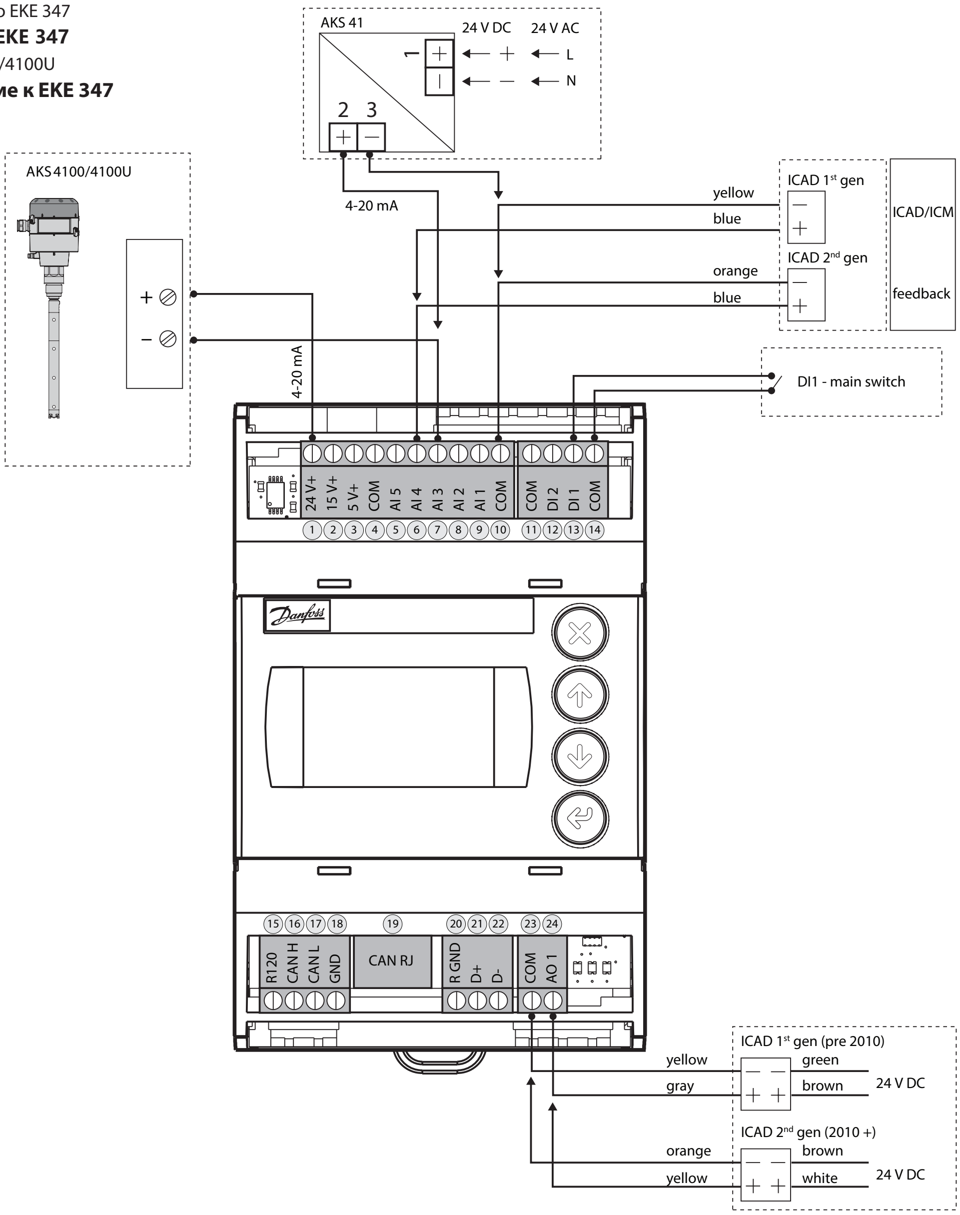
AKS 41/41U から AKS 4100/4100U
AC 電源 AKS 41/41U から DC 電源
AKS 4100/4100U へ

Замена AKS 41 на AKS 4100
AKS 41/41U перененный ток на
AKS 4100/4100U постоянный ток



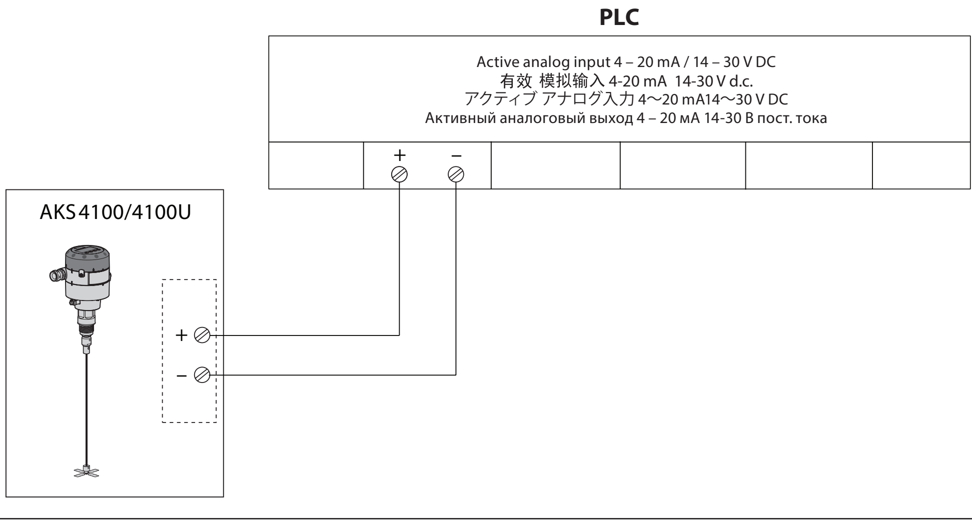
9

AKS 4100/4100U connected to EKE 347
AKS 4100/4100U 连接到 EKE 347
EKE 347 に接続した AKS 4100/4100U
AKS 4100 подсоединение к EKE 347



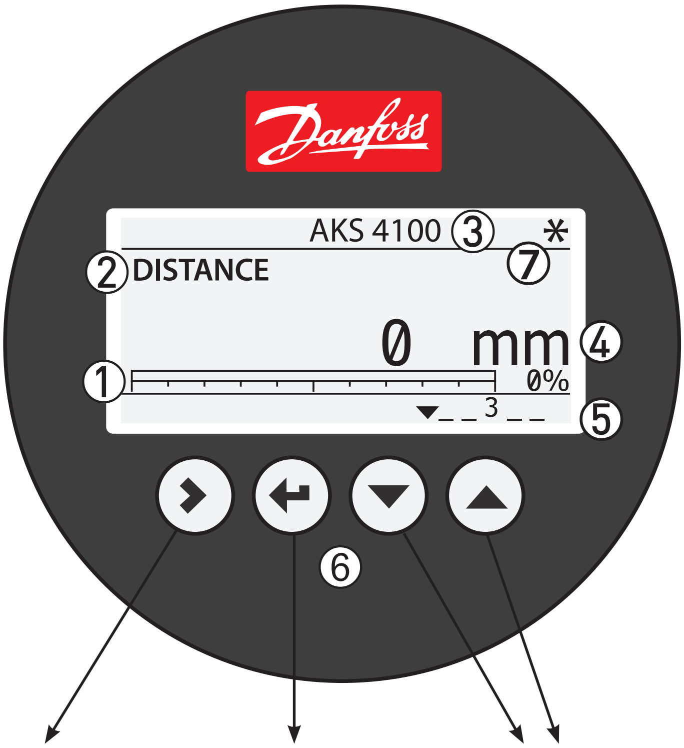
10

AKS 4100/4100U connected to PLC
AKS 4100/4100U 连接到 PLC
PLC に接続した AKS 4100/4100U
AKS 4100 подсоединение к PLC



11

en



Enter menu system
Enter QUICK SETUP

Unit change at distance/level readout:
m, cm, mm, in, ft

Change between:
Distance*
Level**
Output (%)***
Output (mA)****

12

① 4 – 20 mA output displayed as bar graph and in percentage [%]
② Measurement name (in this example, DISTANCE)
③ Device tag name
④ Measurement reading and unit
⑤ Device status (markers)
Marker 1, 2 and 3 (Error)
Hardware problem; the Signal Converter hardware is defective. Contact Danfoss.
Marker 4 and 5 (Notification)
Depending on the level, the marker is ON or OFF. Used for Danfoss service information only.
⑥ Keypad buttons
⑦ Flashing star indicating unit in operation.

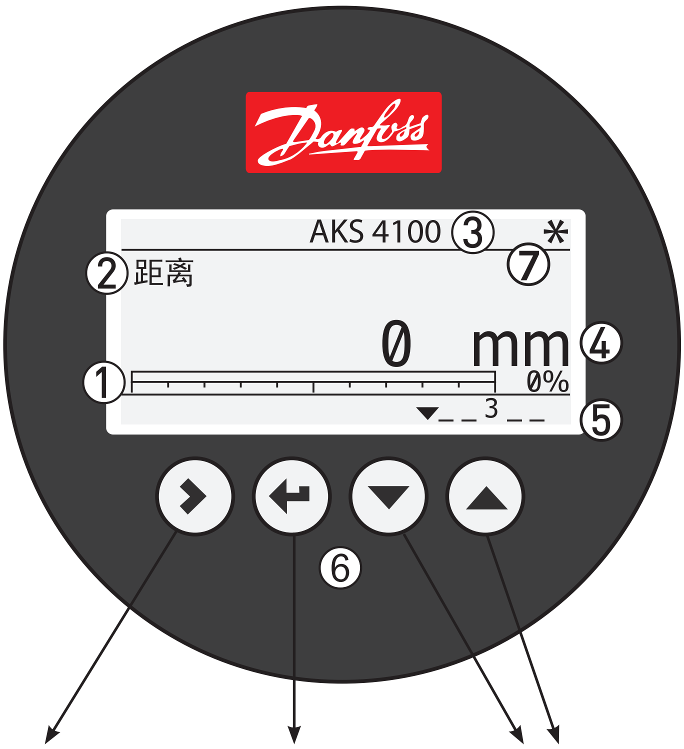
* DISTANCE is a display option.
If the display is set to “DISTANCE” the displayed value will be the distance from the Reference point to the top surface of the liquid refrigerant (see fig. 5).

** LEVEL is display option.
If the display is set to “LEVEL” then the value displayed will be:
PROBE LENGTH (entered in QUICK SETUP)
– DISTANCE (see fig. 5).

*** OUTPUT (%) is display option.
Will represent the level of refrigerant,in percent, scaled (entered in QUICK SETUP) according to: SCALE 4 mA (0%), SCALE 20 mA (100%) (see fig. 5).

**** OUTPUT I (mA) is display option.
Will represent the level of refrigerant,in 4-20 milliampere, scaled (entered in QUICK SETUP) according to: SCALE 4 mA (4 mA), SCALE 20 mA (20 mA) (see fig. 5).

zh



进入Menue system（菜单系统）
进入 QUICK SETUP（快速设置）

距离/液位读数的单位设置：
m、cm、mm、in、ft

切换选项：
Distance（距离） *
Level（液位）液位**
Output（输出）(%)***
Output（输出）(mA)****

12

① 4 – 20 mA 输出（以柱状图和百分比[%]显示）
② 测量值名称(本例中为距离)
③ 设备名称
④ 测量读数和单位
⑤ 设备状态（标记）
标记1、2和3（错误）
硬件问题；信号转换器硬件有缺陷。请联系丹佛斯。
标记4和5（通知）
根据不同的液位，标记为开或关。只用于丹佛斯服务信息。
⑥ 按键
⑦ 闪烁的星星，表示设备正在工作。

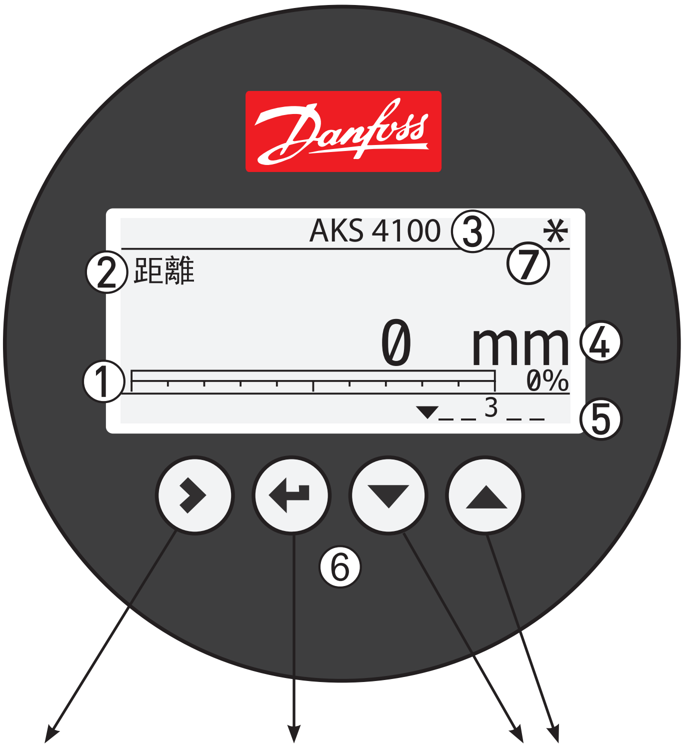
* “Distance（距离）”是一个显示选项。
若将显示设为“Distance”，则显示的值是参照点至制冷剂液面顶部的距离（参考图 5）。

** “Level（液位）”是一个显示选项。
若将显示设为“Level”，则显示的值为：
探头长度（在快速设置中输入） – Level（参考图 5）。

*** “Output（输出）(%)”是一个显示选项。
将以百分比形式显示制冷剂的液位，根据下列条件确定标度（在快速设置中输入）：
SCALE 4 mA(0%)，SCALE 20 mA(100%)（参考图5）。

**** “Output（输出）(mA)”是一个显示选项。
将以 4-20 毫安显示制冷剂的液位，根据下列条件确定标度（在快速设置中输入）：
SCALE 4 mA(4 mA)，SCALE 20 mA (20 mA)（参考图 5）。

12



メニュー システムに移動 クイックセットアップ に移動

距離/レベル読み値の単位を変更:
m、cm、mm、インチ、フィート

以下のいずれかに変更:
距離*
レベル*
出力 (%)***
出力 (mA)****

① 4 – 20 mA 出力をバーグラフとパーセンテージ [%] で表示

② 計測名 (この例では 距離)

③ 装置タグ名

④ 計測読み値と単位

⑤ 装置ステータス (数値)
数値 1、2、3 (エラー)
ハードウェアの問題; 信号変換器 ハードウェアが故障。Danfoss にご連絡ください。
数値 4 および 5 (通知)
レベルに応じて数値がオンまたはオフになります。Danfos のサービス情報としてのみ使用します。

⑥ キーパッド ボタン

⑦ 点滅する星印は装置が動作中であることを示します。

* 「距離」は表示オプションです。
表示が「距離」に設定されている場合、表示される値は基準点から液体冷媒の上面までの距離になります (図5を参照)。

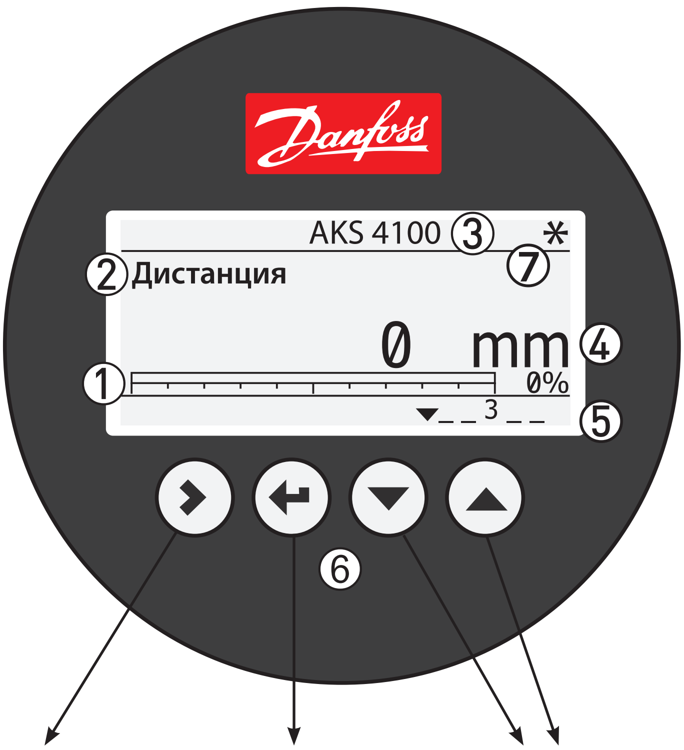
** 「LEVEL "レベル"」は表示オプションです。
表示が「LEVEL "レベル"」に設定されている場合には、表示される値は以下のようになります:プローブ長 (高速設置で入力) – 距離 (図 5 を参照)。

*** 「OUTPUT (%) "出力(%)"」は表示オプションです。冷媒のレベルを (クイックセットアップで入力した) 以下のスケールを基準としたパーセント単位で表示します: 4mA位置 (0%)、20mA位置 (100%) (図 5 を参照)。

**** 「OUTPUT I (mA) "出力 (mA)"」は表示オプションです。冷媒のレベルを (クイックセットアップで入力した) 以下のスケールを基準として 4～20m A の範囲で表示します:4mA位置 (4 mA)、20mA位置 (20 mA) (図 5 を参照)。

ru

12



Вход в меню системы. Запуск режима «Быстрая настройка».

Изменение единиц измерения для отображения расстояния/ уровня: м, см, мм, дюймы, футы.

Переключение между:
Расстояние*
Уровень**
Выходной сигнал (%)***
Выходной сигнал (мА)****

① Выходной сигнал 4 – 20 мА отображается на дисплее ввиде графической шкалы и виде процентов (%).
② Наименование измеряемого параметра (в данном примере это расстояние).
③ Модель уровнимера.
④ Единицы измерения.
⑤ Состояние прибора (в виде цифр):
Цифры 1, 2 и 3 (Ошибка)
Проблема с оборудованием; Отсутствие сигнала; Низкое напряжение. Обратитесь в Danfoss.
Цыфры 4 и 5 (Уведомления)
В зависимости от значения уровня.
⑥ Кнопки панели управления
⑦ Мигающая звёздочка, показывает, что прибор работает.

* РАССТОЯНИЕ.
В случае отображения на дисплее параметра "РАССТОЯНИЕ", измеряемой величиной будет являться расстояние между нулевой точкой и поверхностью жидкого холодильного агента (см. Рис. 5).

** УРОВЕНЬ.
В случае отображения на дисплее параметра "УРОВЕНЬ", измеряемой величиной будет являться разница значений "ДЛИНА ТРОСА" (вводится при "БЫСТРОЙ НАСТРОЙКЕ") – РАССТОЯНИЕ (см. Рис. 5).

*** Выходной сигнал (%).
Указываетпроцент заполнения сосуда холодильным агентом в зависимости от значения:
Точки 4 мА (0%) и Точки 20 мА (100%)
(см. Рис. 5) (указываются при "БЫСТРОЙ НАСТРОЙКЕ").

**** Выходной сигнал (мА).
Указывает уровень холодильного агента в сосуде, в соответствии с диапозоном
4 – 20 мА, в зависимости от значения:
Точки 4 мА (0%) и Точки 20 мА (100%)
(см. Рис. 5) (указываются при "БЫСТРОЙ НАСТРОЙКЕ").

ENGLISH

Please observe that AKS 4100/4100U is intended to always be installed in a standpipe (column / bypass / stilling well). A Standpipe is commonly used when:

- Servicing the AKS 4100
- There is highly conductive foam in the tank.
- The liquid is very turbulent or agitated.

Refrigerants

AKS 4100/4100U is designed specifically to measure liquid level in the most commonly used refrigerants, including R717(ammonia), HCFC, HFC and non corrosive gases / liquids.

AKS 4100/4100U can also be used with R744 (CO_2) in the coaxial version. Please see the technical brochure for further details.

Basic data

AKS 4100/4100U is a passive 2-wired 4 – 20 mA sensor that is loop powered.

Supply Voltage
14 – 30 V DC Min/Max. value for a max. output of 22 mA at the terminal

Load
RL [Ω] ≤ ((Uext -14 V)/20 mA).
– Default (Error output set to 3.6 mA)
RL [Ω] ≤ ((Uext -14 V)/22 mA).
– (Error output set to 22 mA)

Cable gland
AKS 4100 PG 13, M20×1.5 ; (cable diameter: 6-8 mm (0.24-0.31in.)
AKS 4100U ½ in. NPT

Terminals (spring loaded)
0.5-1.5 mm^2 (~20-15 AWG)

Enclosure
IP 67 (~NEMA type 4X)

Refrigerant temperature
-60 – 100 °C / -76 – 212 °F

Refrigerants
The listed refrigerants are qualified and approved by Danfoss:

R717 / NH_3:	-40 – 50 °C / -40 – 122 °F
R744 / CO_2:	-50 – 15 °C / -58 – 59 °F
HCFC:	
R22:	-50 – 48 °C / -58 – 118 °F
HFC:	
R404A:	-50 – 15 °C / -58 – 59 °F
R410A:	-50 – 15 °C / -58 – 59 °F
R134A	-40 – 50 °C / -40 – 122 °F

(Further details in the Technical Brochure)

Ambient temperature
-40 – 80 °C / -40 – 175 °F
For HMI : -20 – 60 °C / -4 – 140 °F

Process pressure
-1 – 100 barg / -14.5 – 1450 psig

Mechanical process connection with 5 m / 197 in. Ø2 mm / 0.08 in. stainless cable:
AKS 4100 G1 inch pipe thread. Aluminium gasket included
AKS 4100U ¾ in. NPT

(Further details in the Technical Brochure)

Mechanical Installation

Preparations prior to Mechanical Installation
Disassemble the Signal Converter from the Mechanical process connection (use 5´mm hex key, **see fig. 6**). Fit the red protection cover on top of the Mechanical process connection to protect it againt any moisture or dirt paticles.

Content supplied (fig. 1)
① Signal Converter (with or without HMI)
② Mechanical process connection with 5 m / 197 in. Ø2 mm / 0.08 in. stainless wire
③ Counterweight
④ Accessory bag comprising:
3 mm set screws.
Red cover to protect mechanical process connection ② prior to mounting Signal converter.
Setting label.

Note:
Stand pipe design guidelines:
The side connection pipes must NOT penetrate into the stand pipe (**fig. 2a**)
Recommended diameter of the side connection pipe: 0.5 x stand pipe diameter (e.g. if stand pipe has diameter DN100, the side connection must be diameter DN50 or smaller) (**fig. 2a**).

If above design guidelines for side connection are not fulfilled, one of the following options are recommended:

1. Increase Detection Delay. Parameter 2.3.6.
   We recommend to increase the Detection Delay (parameter 2.3.6) from the standard 0 mm to a value below the lowest point of the top side connection plus 50 mm (**fig. 2a**).

   Changing the Detection Delay (parameter 2.3.6) does not require changing the (0%) 4mA and (100%)20mA settings in the AKS 4100/4100U Within the Detection Delay zone, no measuring will take place.
2. Exchange from Cable to Coaxial version.

The stand pipe must have the SAME diameter through out the entire length. If standpipe diameter differs in width (**fig. 2b**) the Cable version is not recommended. Coaxial version should be used.

Adjustment of the counterweight blades
Allow 5 mm space between the guided blades and the inner wall of the pipe **(see fig. 2c)**. Use side cutters to trim the guided blades to fit the actual standpipe diameter **(see fig. 3)**.

Adjustment of the cable probe

Please observe that the stainless steel wire is not permanently creased or kinked.

Always use the reference point, at the **Mechanical Process Connection (see fig. 4)**, as a starting point for all measuring to determine:

- Where to cut the cable.
- Probe length **(see fig. 5)**
- Scale 4 mA **(see fig. 5)**
- Scale 20 mA **(see fig. 5)**

Note the probe length, Scale 4 mA and Scale 20 mA for use later when programming the HMI (Human Machine Interface) on the AKS 4100/4100U.

Follow these instructions and **see fig. 4 & 5**:

1. Measure the inner length of the Standpipe.
2. *Preparation before cutting the cable*
   Known data:
   Space below counterweight: 20 mm / 0.8 in.
   Steel wire insertion length in counterweight: 12 mm / 0.5 in.
   counterweight height: 33 mm / 1.3 in.

   Max Probe length =
   Standpipe inner length
   – Space below counterweight (20 mm / 0.8 in.)

   The cable length =
   Max probe length
   \+ Steel wire insertion length in counterweight (12 mm / 0.5 in.)
   – Counterweight height (33 mm / 1.3 in.)
3. Measure out the cutting point of the cable. Measure from the reference point **(fig. 4)** and cut the cable.
4. Fit the counterweight on the cable and secure the two set screws with a 3 mm Allen Key **(fig. 3)**.
5. Lower the counterweight down through the threaded hole. **Make sure that the counterweight is gliding down through the pipe without any resistance and that the cable is straight (not touching the inner walls of the stand pibe or any incoming piping (see fig. 2a)).**
6. Use a torque wrench to tighten the mechanical process connection **(fig. 1, item 2)** to 120 Nm (89 lb/ft).

Calculating the measuring range

***4 mA** setting for max. measuring range:*
= Max probe length
– Counterweight height (33 mm / 1.3 in.)
– Bottom dead zone (see **fig. 5**)

***20 mA** setting for max. measuring range:*
= Top dead zone (see **fig. 5**)

Example
Known data:
Space below counterweight: 20 mm / 0.8 in.
Steel wire insertion length in counterweight: 12 mm / 0.5 in.
counterweight height: 33 mm / 1.3 in.
Preconditions:
Factory setting is used
Refrigerant = Ammonia
Standpipe inner length = 3100 mm / 122 in.
Max probe length =
3100 mm – 20 mm = 3080 mm
(122 in. – 0.8 in. = 121.3 in.)

The cable length:
Max probe length =
\+ Steel wire insertion length in counterweight (12 mm / 0.5 in.)
– Counterweight height (33 mm / 1.3 in.)
3080 mm + 12 mm – 33 mm = **3059 mm**
(121.3 in. + 0.5 in. – 1.3 in. = **120.4 in.)**

***4 mA** Setting for Max. Measuring Range:*
Max probe length (3080 mm / 121.3 in.)
– Counterweight height (33 mm / 1.3 in.)
– Bottom dead zone (see fig. 5)
(210 mm / 8.3 in.) = **2837 mm / 111.7 in.**

***20 mA** Setting for Max. Measuring Range:*
= Top dead zone (see fig. 5) = 120 mm / 4.7 in.

How to mount the AKS 4100/4100U Converter (see fig 6)

1. Unscrew the set and ventilation screws with a 5 mm Hexagon key in the Signal converter.
2. Push the Signal Converter downwards to stop on the Mechanical process connection
3. Turn the Signal Converter to the wanted position.
4. Screw the set screw with a 5 mm Hexagon key.
5. Screw the ventilation screw with a 5 mm Hexagon key.

Electrical installation/connection

Output terminals (fig. 7 and 8):
1. Current output –
2. Current output +
3. Grounding terminal

Electrical installation procedure
1. Use a 2.5 mm Allen wrench to loosen the cover stop.
2. Remove the terminal compartment cover from the housing.
3. Do not disconnect the wire from the terminal compartment cover.

Put the terminal compartment cover adjacent to the housing.

4. Connect the wires to the device. Tighten the cable entry glands.
5. Attach the terminal compartment cover to the housing.
6. Use a 2.5 mm Allen wrench to tighten the cover stop.

Start up:
- Connect the converter to the power supply.
- Energize the converter.

Devices with the HMI display option only: After 10 seconds the screen will display "Starting up". After 20 seconds the screen will display the software version numbers. After 30 seconds the default screen **(fig. 12)** will appear.

Precautions when changing from AKS 41/41U to AKS 4100/4100U

Note:
AKS 41/41U supports both AC and DC supply whereas the AKS 4100/4100U is using DC supply only. **Following the instructions in fig. 9.**

Connecting to controller or PLC
Follow the instructions in **fig. 10 or 11**.

The current output will be set to 3.6 mA whenever the AKS 4100/4100 detects an error like Marker 1, 2 or 3 (see page 4).

Quick Setup →

Note:
The signal converter can be programmed with or without mechanical process connector assembled.

Quick Setup (all values below are only examples)

- Connect the device to the power supply (see the section "Electrical installation/ connection".
- Press 3 times.

AKS 4100
QUICK SETUP?
YES NO

- Press

AKS 4100
PROBETYPE
SINGLECABLE

- Press or to select between SINGLE, COAXIAL D14 and COAXIAL D22.
Choose **SINGLE** and press to confirm.

AKS 4100
PROBE LENGTH
05000 mm

- Press to change the PROBE LENGTH.
Press to change the position of the cursor.
Press to decrease the value or to increase the value.
Press to confirm.

AKS 4100
SCALE 4mA
04946 mm

- Press to change of SCALE 4 mA.
Press to change the cursor position.
Press to decrease the value or to increase the value.
Press to confirm.

AKS 4100
SCALE 20mA
00070 mm

- Press to change of SCALE 20 mA.
Press to change the cursor position.
Press to decrease the value or to increase the value.
Press to confirm.

AKS 4100
QUICK SETUP
COMPLETED IN 8

- Wait for QUICK SETUP to complete 8-second timeout

AKS 4100
1.0.0
QUICK SETUP

- Press to confirm.

AKS 4100
1.0.0
STORE NO

- Press or to select either STORE NO or STORE YES.
Press to confirm.

Default screen appears:

AKS 4100
DISTANCE
5000 mm

Quick Setup completed

You have the possibility of checking your settings by pressing twice.

AKS 4100	
SINGLE CABLE	5000 mm
(0%) 4 mA	4877 mm
(100%) 20 mA	120 mm

Press to return to default screen.

How to force mA output (all values below are only examples)

Default screen

AKS 4100
DISTANCE
5000 mm

- Press

AKS 4100
1.0.0
QUICK SETUP

- Press

AKS 4100
2.0.0
SUPERVISOR

- Press

AKS 4100
2.0.0
____

Enter password:

AKS 4100
2.1.0
INFORMATION

- Press

AKS 4100
2.2.0
TESTS

- Press

AKS 4100
2.2.1
SET OUTPUT

- Press

AKS 4100
SET OUTPUT
3.5 mA

- Press to decrease the value or to increase the value.
Press to confirm.

AKS 4100
SET OUTPUT
8 mA

- Press 4 times to return to default screen.

Default screen appears:

AKS 4100
DISTANCE
5000 mm

Force mA completed and disabled

Optional Procedure

If the temperature condition in the stand pipe is known, a constant (dielectric constant of the refrigerant gas) **can be** entered (parameter 2.5.3 GAS EPS.R), in order to obtain lower Top and Bottom Dead Zone values **(see fig. 5)**.

How to enter dielectric constant of refrigerant gas (all values below are only examples)

Default screen

```
AKS 4100
DISTANCE
5000 mm
```

- Press ►

```
AKS 4100
1.0.0
QUICK SETUP
```

- Press ▲

```
AKS 4100
2.0.0
SUPERVISOR
```

- Press ►

```
AKS 4100
2.0.0
________
```

Enter password:
► ↵ ▼ ▲ ► ↵

```
AKS 4100
2.1.0
INFORMATION
```

- Press ▲ 4 times.

```
AKS 4100
2.5.0
APPLICATION
```

- Press ►

```
AKS 4100
2.5.1
TRACING VEL.
```

- Press ▲ 2 times.

```
AKS 4100
2.5.3
GAS EPS. R
```

- Press ► to check/change GAS EPS.R. (Select the correct value from the tables below and on page 8)
  Press ► to change cursor-position.
  Press ▼ to decrease the value or ▲ to increase the value.

```
AKS 4100
GAS EPS. R
1.066
```

- Press ↵ to confirm.

```
AKS 4100
2.5.3
GAS EPS. R
```

- Press ↵ 3 times.

```
AKS 4100
1.0.0
STORE NO
```

- Press ▼ or ▲ to select between STORE NO or STORE YES.
  Select STORE YES by pressing ↵

  Default screen appears:

```
AKS 4100
DISTANCE
5000 mm
```

Entering the dielectric constant of refrigerant gas completed

Saturated vapour dielectric constant (default value: 1.066)

R717 (NH_3)

Temperature range:
-60 – 50 °C / -76 – 122 °F

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1.00
-41 – -18	42 – 0	1.01
-17 – -5	1 – 23	1.02
-4 – 4	24 – 39	1.03
5 – 12	40 – 54	1.04
13 – 18	55 – 64	1.05
19 – 24	65 – 75	1.06
25 – 28	76 – 82	1.07
29 – 33	83 – 91	1.08
34 – 37	92 – 99	1.09
38 – 40	100 – 104	1.10
41 – 44	105 – 111	1.11
45 – 47	112 – 117	1.12
48 – 50	118 – 122	1.13

R22

Temperature range:
-60 – 48 °C / -76 – 118 °F

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -50	-76 – -58	1.00
-49 – -25	57 – -13	1.01
-24 – -10	-12 – 14	1.02
-9 – 0	15 – 32	1.03
1 – 8	33 – 46	1.04
9 – 15	47 – 59	1.05
16 – 21	60 – 70	1.06
22 – 26	71 – 79	1.07
27 – 31	80 – 88	1.08
32 – 35	89 – 95	1.09
36 – 39	96 – 102	1.10
40 – 42	103 – 108	1.11
43 – 45	109 – 113	1.12
46 – 48	114 – 118	1.13

R744 (CO_2)

Temperature range:
-56 – 15 °C / -69 – 59 °F

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-56.0 – -42.0	-69 – -43	1.01
-41.0 – -28.0	-42 – -18	1.02
-27.0 – -17.0	-17 – 2	1.03
-16.0 – -9.0	3 – 16	1.04
-8.0 – -3.0	17 – 27	1.05
-2.0 – 2	28 – 36	1.06
3 – 7	37 – 45	1.07
8 – 11	46 – 52	1.08
12 – 14	53 – 58	1.09
15	59	1.10

R134a

Temperature range:
-60 – 50 °C / -76 – 122 °F

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1.00
-41 – -18	-42 – -0	1.01
-17 – -4	1 – 25	1.02
-3 – 5	26 – 41	1.03
6 – 13	42 – 56	1.04
14 – 20	57 – 68	1.05
21 – 25	69 – 77	1.06
26 – 30	78 – 86	1.07
31 – 34	87 – 94	1.08
35 – 38	95 – 100	1.09
39 – 42	101 – 108	1.10
43 – 45	109 – 113	1.11
46 – 48	114 – 119	1.12
49 – 50	120 – 122	1.13

Saturated vapour dielectric constant

R410A
Temperature range:
-65 – 15 °C / -85 – 59 °F

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-65 – -47	-85 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -13	-1 – 9	1.05
-12 – -8	10 – 18	1.06
-7 – -4	19 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 4	33 – 40	1.09
5 – 7	41 – 45	1.10
8 – 10	46 – 50	1.11
11 – 12	51 – 54	1.12
13 – 15	55 – 59	1.13

R507
Temperature range:
-60 – 15 °C / -76 – 59 °F

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -48	-76 – -54	1.01
-47 – -36	-53 – -32	1.02
-35 – -28	-31 – -18	1.03
-27 – -21	-17 – -6	1.04
-20 – -15	-17 – -5	1.05
-14 – -10	-4 – 14	1.06
-9 – -6	13 – 22	1.07
-5 – -2	23 – 29	1.08
-1 – 2	30 – 36	1.09
3 – 5	37 – 41	1.10
6 – 8	42 – 47	1.11
9 – 11	48 – 52	1.12
12 – 13	53 – 56	1.13
14 – 15	57 – 59	1.14

R404A
Temperature range:
-60 – 15 °C / -76 – 59 °F

Temperature [°C]	Temperature [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -47	-76 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -14	-1 – 7	1.05
-13 – -9	8 – 16	1.06
-8 – -4	17 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 3	33 – 38	1.09
4 – 6	39 – 43	1.10
7 – 9	44 – 49	1.11
10 – 12	50 – 54	1.12
13 – 15	55 – 59	1.13

How to change the language setting (Default: English)

Default screen

AKS 4100
DISTANCE
5000 mm

- Press (›)

AKS 4100
1.0.0
QUICK SETUP

- Press (▲)

AKS 4100
2.0.0
SUPERVISOR

- Press (›)

AKS 4100
2.0.0
____

Enter password:
(›) (↵) (▼) (▲) (›) (↵)

AKS 4100
2.1.0
INFORMATION

- Press (▲) 6 times

AKS 4100
2.7.0
DISPLAY

- Press (›)

AKS 4100
2.7.1
LANGUAGE

- Press (›)

AKS 4100
LANGUAGE
ENGLISH

- Press (▼) or (▲) to see the language possibilities
  Press (↵) to confirm.

AKS 4100
2.7.1
LANGUAGE

- Press (↵) 3 times

AKS 4100
1.0.0
STORE NO

- Press (▼) or (▲) to select between STORE NO or STORE YES.
  Select STORE YES by pressing (↵)
  Default screen appears:

AKS 4100
DISTANCE
5000 mm

Language setup completed

Reset to factory setting

- Go to SUPERVISOR menu (see page 7).
- Go to parameter 2.9.4 Reset Factory.
- Select RESET FACTORY YES
- Press (↵) 3 times to return to default screen.

Factory reset completed.

中文

请注意，AKS 4100/4100U 总是安装在立管中。立管通常应用于下列情况：
- 检修 AKS 4100
- 容器中存在高传导泡沫
- 液体处于紊流或翻腾状态

制冷剂

AKS 4100/4100U 专用于测量常用制冷剂的液位，包括氨、HCFC、HFC 和非腐蚀性气体/液体。

同轴版本的 AKS 4100/4100U 还适用于 R744（二氧化碳）。详情请参考技术手册。

基本参数

AKS 4100/4100U 是一款回路供电的无源二线制 4 – 20 mA 传感器。

供电电压
14 – 30 V DC最小/最大值
端子最大输出为22 mA

负载
RL［Ω］≤((Uext -14 V)/20 mA)
– 默认（差错输出设为 3.6 mA）
RL［Ω］≤((Uext -14 V)/22 mA)
– 默认（差错输出设为 22 mA）

电缆封套
AKS 4100　PG 13, M20×1.5；
(电缆直径：6-8 mm(0.24-0.31in.)
AKS 4100U　½ in. NPT

接线端子（弹簧承载）
0.5-1.5 mm² (~20-15 AWG)

外壳
IP 67(~NEMA 4X)
制冷剂温度
-60 – 100 °C / -76 – 212 °F

制冷剂
丹佛斯认可适用的制冷剂：

R717 / NH_3:	-40 – 50 °C / -40 – 122 °F
R744 / CO_2:	-50 – 15 °C / -58 – 59 °F
HCFC:	
R22:	-50 – 48 °C / -58 – 118 °F
HFC:	
R404A:	-50 – 15 °C / -58 – 59 °F
R410A:	-50 – 15 °C / -58 – 59 °F
R134A	-40 – 50 °C / -40 – 122 °F

(详情请参阅《技术手册》)

环境温度
-40 – 80 °C / -40 – 175 °F
HMI：-20 – 60 °C / -4 – 140 °F

操作压力
-1 – 100 barg / -14.5 – 1450 psig
机械连接件，带 5 m / 197 in.Ø2 mm / 0.08 in.不锈钢缆：
AKS 4100　1″ G 螺纹
自带铝制垫圈
AKS 4100U　¾ in. NPT

（详情请参阅《技术手册》）

机械安装

机械安装前的准备
从机械连接件上卸下信号转换器（用 5’mm 六角扳手，参阅图6）。将红色保护盖安装到机械连接件的顶部，使之免受潮气或污垢的影响。

装箱清单（图1）
① 信号转换器（搭配/不搭配 HMI）

② 机械连接件，带 5 m / 197 in.
Ø2 mm / 0.08 in.不锈钢线

③ 平衡坠子
④ 附件袋内有：
3 mm 固定螺钉
安装信号转换器前用于保护机械连连接件②的红色保护盖设置标签

注意：
立管设计指导：

旁通管不能穿透到立管（如图2a）。推荐旁通管直径：0.5倍立管直径（例：如果立管直径为DN100，旁通管直径必须小于或等于DN50）（如图2a）。

如果上述旁通管设计指导不能满足，推荐以下选项之一：

1、增加检测延迟——参数2.3.6
建议增加检测延迟（参数2.3.6）从标准0 mm到低于上部旁通管最低点加50 mm的某个值（如图2）。

改变检测延迟（参数2.3.6）不需要改变AKS 4100/4100U关于(0%) 4 mA和(100%)20 mA的参数设置。在检测延迟区，没有测量发生。

2、更换为同轴版本

整条立管必须保持相同的直径。如果立管直径变化（如图2b）不推荐使用线缆版本，建议使用同轴版本。

调整平衡叶片
导叶与管道内壁之间应有 5 mm 间距（参阅图2c）。剪切导叶，使之与立管的实际直径相吻合（参阅图3）。

调整电缆探头

切勿让不锈钢丝
出现永久性折痕或扭结。

开始任何测量时，应当利用机械连接件（参阅图4）上的参照点，以便确定：
- 从何处切割线缆
- 探头长度（参阅图5）
- 标度 4 mA （参阅图5）
- 标度 20 mA （参阅图5）

注意探头长度，标度 4 mA 和标度 20 mA 用于稍后对 AKS 4100/4100U 上的 HMI（人机界面）进行编程。

按照下列说明操作，同时参阅图4和5：
1. 测量立管内部长度

2. 切割线缆前的准备
已知数据：
平衡坠子下方空间：20 mm / 0.8 in.
钢缆插入平衡重的长度：12 mm / 0.5 in.
平衡坠子高度：33 mm / 1.3 in.

最大探头长度 =立管内部长度
– 平衡坠子下方空间
20 mm / 0.8 in.

线缆长度 =最大探头长度+ 钢线插入平衡坠子的长度 12 mm / 0.5 in.
– 平衡坠子高度 33 mm / 1.3 in.

3. 测得线缆的切割点
从参照点开始测量（图4）并切割线缆

4. 将平衡坠子安装到线缆上，并用 3 mm 内六角扳手拧紧两颗固定螺钉（图3）

5. 通过螺纹孔调低平衡坠子的位置。确保平衡坠子沿着管道无阻力下滑，同时线缆保持平直（不接触内管或接入管道的内壁参阅图 2a)

6. 用扭力扳手拧紧机械连接件（图1，第2项），力矩为 120 Nm(89 lb/ft)

计算测量范围
4 mA 设置最大测量范围：
= 最大探头长度
– 平衡重高度33 mm(1.3 in.)
– 底端死区（参阅图5）

20 mA 设置最大测量范围：
= 顶端死区（参阅图5）

示例
已知数据：
平衡坠子下方空间：20 mm(0.8 in.)
钢线插入平衡坠子的长度：12 mm(0.5 in.)
平衡坠子高度：33 mm(1.3 in.)
前提条件：
使用出厂设定值
制冷剂 = 氨立管内部长度 = 3100 mm(122 in.)
最大探头长度 =3100 mm – 20 mm = 3080 mm
*122 in. – 0.8 in. = 121.3 in.)

线缆长度：
最大探头长度 =
+ 钢线插入平衡坠子的长度12 mm / 0.5 in.
– 平衡坠子高度33 mm(1.3 in.)
3080 mm + 12 mm – 33 mm = <u>3059 mm</u>
(121.3 in. + 0.5 in. – 1.3 in. = <u>120.4 in.</u>)

4 mA 设置最大测量范围：
最大探头长度3080 mm(121.3 in.)
– 平衡坠子高度(33 mm(1.3 in.))
– 底端死区(参阅图 5)
(210 mm(8.3 in.)= <u>2837 mm(111.7 in.)</u>

20 mA 设置（最大测量范围：
= 顶端死区（参阅图 5）= 120 mm(4.7 in.)

如何安装 AKS 4100/4100U 转换器（参阅图 6）
1. 使用信号转换器里面的 5mm 内六角扳手拧开固定螺栓与通风螺栓。
2. 向下推动信号转换器，直至它停在机械连接件上。
3. 转动信号转换器至想要的位置。
4. 用 5 mm 内六角扳手拧紧固定螺栓。
5. 用 5 mm 内六角扳手拧紧通风螺栓。

电气安装/连接

输出端子（图 7 和 8）：
1.电流输出 –
2.电流输出 +
3.接地端

电气安装步骤
1. 用 2.5 mm 内六角扳手松开盖挡。
2. 从机壳上取下端子舱盖。
3. 不要断开钢缆与端子舱盖的连接。将端子舱盖放在机壳旁边。
4. 将钢缆连接到设备。上紧线缆入口封套。
5. 将端子舱盖安装到机壳上。
6. 用 2.5 mm 内六角扳手拧紧盖挡。

启动
- 将转换器连接到电源。
- 将转换器通电。

若设备带选配的 HMI 显示屏则仅在：10秒后屏幕将显示“Starting up（启动）”。20秒后屏幕将显示软件版本号。30秒后显示默认屏幕（图12）。

将 AKS 41/41U 改为 AKS 4100/4100U 时需要注意。

注意：
AKS 41/41U 支持交流和直流电源，而 AKS 4100/4100U 只能使用直流电源。按照图9的说明进行操作。

连接到控制器或 PLC
按照图10或11的说明进行操作。

注意：
AKS 4100/4100 检测到错误，电流输出就会被设为 3.6 mA，设备状态将标记为 1、2、3

注意：
无论是否装有机械连接件，均可对信号转换器进行编程。

快速设置（下面所有数值均为举例）

- 将设备连接到电源（参阅“电气安装/连接”）。
- 连按 ⊙ 3 次。

AKS 4100
快速设置
是 否

- 按下 ⊙

AKS 4100
探头类型
线缆式

- 按下 ⊙ 或 ⊙ ，，在SINGLE（电缆形式），COAXIAL （同轴形式）D14或COAXIAL （同轴形式）D22之间选择一项。
  按下 ⊙ 以示确认。

AKS 4100
探头长度
05000 mm

- 按下 ⊙ 更改探头长度。
  按下 ⊙ 更改游标位置。
  按下 ⊙ 调低数值，或者按下 ⊙ 调高数值。
  按下 ⊙ 以示确认。

AKS 4100
4mA对应量程
04946 mm

- 按下 ⊙ 更改SCALE（标度）4 mA。
  按下 ⊙ 更改游标位置。
  按下 ⊙ 调低数值，或者按 ⊙ 调高数值。
  按下 ⊙ 以示确认。

AKS 4100
20mA对应量程
00070 mm

- 按下 ⊙ 更改SCALE（标度） 20 mA。
  按下 ⊙ 更改游标位置。
  按下 ⊙ 调低数值，或者按 ⊙ 调高数值。
  按下 ⊙ 以示确认。

AKS 4100
快速设置
完成 8

- 等待8秒，快速设置完成

AKS 4100
1.0.0
快速设置

- 按下 ⊙ 以示确认。

AKS 4100
1.0.0
保存设置 NO

- 按下 ⊙ 或 ⊙ ，在 STORE NO （保存否）和STORE YES（保存是） 中选择一项。
  按下 ⊙ 以示确认。

  显示默认屏幕：

AKS 4100
距离
5000 mm

快速设置完成

您可以检查
您的设置，方法是连按两次 ⊙ 。

AKS 4100	
线缆式	5000 mm
(0%) 4 mA	4877 mm
(100%) 20 mA	120 mm

按下 ⊙ ⊙ ⊙ 返回默认屏幕。

强制 mA 输出（下面所有数值均为举例）

默认屏幕

AKS 4100
距离
5000 mm

- 按下 ⊙

AKS 4100
1.0.0
快速设置

- 按下 ⊙

AKS 4100
2.0.0
操作员

- 按下 ⊙

AKS 4100
2.0.0
____

输入密码：
⊙ ⊙ ⊙ ⊙ ⊙ ⊙

AKS 4100
2.1.0
信息

- 按下 ⊙

AKS 4100
2.2.0
测试

- 按下 ⊙

AKS 4100
2.2.1
测试电流

- 按下 ⊙

AKS 4100
测试电流
3.5 mA

- 按下 ⊙ 调高数值，或者按下 ⊙ 调低数值。
  按下 ⊙ 以示确认。

AKS 4100
测试电流
8 mA

- 连按 4次 ⊙ 返回默认屏幕。

默认屏幕

AKS 4100
距离
5000 mm

强制 mA 完成并禁用

可选步骤
若已知立管的温度条件，可以输入一个常数（制冷剂气体的介电常数）（参数 2.5.3 GAS EPS.R），以降低顶端和底端死区的数值（参阅图5）。

如何输入制冷剂气体的介电常数（以下所有数值均为举例）

默认屏幕

AKS 4100
距离
5000 mm

- 按下 ▶

AKS 4100
1.0.0
快速设置

- 按下 ▲

AKS 4100
2.0.0
操作员

- 按下 ▶

AKS 4100
2.0.0
____

输入密码：

▶ ↵ ▼ ▲ ▶ ↵

AKS 4100
2.1.0
信息

- 按下 ▲ 4次

AKS 4100
2.5.0
工况参数

- 按下 ▶

AKS 4100
2.5.1
追従速度

- 按下 ▲ 2次

AKS 4100
2.5.3
气相Er

- 按下 ▶，以便检查/修改 GASEPS.R（气体介电常数）（从下表和第8页的表格中选择正确的数值）
按下 ▶ 更改游标位置。
按下 ▼ 调低数值，或者按下 ▲ 调高数值。

AKS 4100
气相Er
1.066

- 按下 ↵ 以示确认。

AKS 4100
2.5.3
气相Er

- 按下 ↵ 3次。

AKS 4100
1.0.0
保存设置 NO

- 按下 ▼ 或 ▲ ，在 STORE NO（保存 否）和 STORE NO（保存 是）之间进行选择。
按下 ↵ 选择 STORE NO（保存 是）

默认屏幕

AKS 4100
距离
5000 mm

介电常数输入完成。

饱和蒸气的介电常数（默认值：1.066）

R717 （氨）
温度范围：
-60 – 50 °C / -76 – 122 °F

温度 [℃]	温度 [℉]	制冷剂气体的介电常数 参数 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1.00
-41 – -18	42 – 0	1.01
-17 – -5	1 – 23	1.02
-4 – 4	24 – 39	1.03
5 – 12	40 – 54	1.04
13 – 18	55 – 64	1.05
19 – 24	65 – 75	1.06
25 – 28	76 – 82	1.07
29 – 33	83 – 91	1.08
34 – 37	92 – 99	1.09
38 – 40	100 – 104	1.10
41 – 44	105 – 111	1.11
45 – 47	112 – 117	1.12
48 – 50	118 – 122	1.13

R744 （二氧化碳）
温度范围：
-56 – 15 °C / -69 – 59 °F

温度 [℃]	温度 [℉]	制冷剂气体的介电常数 参数 2.5.3 GAS EPS.R
-56.0 – -42.0	-69 – -43	1.01
-41.0 – -28.0	-42 – -18	1.02
-27.0 – -17.0	-17 – 2	1.03
-16.0 – -9.0	3 – 16	1.04
-8.0 – -3.0	17 – 27	1.05
-2.0 – 2	28 – 36	1.06
3 – 7	37 – 45	1.07
8 – 11	46 – 52	1.08
12 – 14	53 – 58	1.09
15	59	1.10

R22
温度范围：
-60 – 48 °C / -76 – 118 °F

温度 [℃]	温度 [℉]	制冷剂气体的介电常数 参数 2.5.3 GAS EPS.R
-60 – -50	-76 – -58	1.00
-49 – -25	57 – -13	1.01
-24 – -10	-12 – 14	1.02
-9 – 0	15 – 32	1.03
1 – 8	33 – 46	1.04
9 – 15	47 – 59	1.05
16 – 21	60 – 70	1.06
22 – 26	71 – 79	1.07
27 – 31	80 – 88	1.08
32 – 35	89 – 95	1.09
36 – 39	96 – 102	1.10
40 – 42	103 – 108	1.11
43 – 45	109 – 113	1.12
46 – 48	114 – 118	1.13

R134a
温度范围：
-60 – 50 °C / -76 – 122 °F

温度 [℃]	温度 [℉]	制冷剂气体的介电常数 参数 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1.00
-41 – -18	-42 – -0	1.01
-17 – -4	1 – 25	1.02
-3 – 5	26 – 41	1.03
6 – 13	42 – 56	1.04
14 – 20	57 – 68	1.05
21 – 25	69 – 77	1.06
26 – 30	78 – 86	1.07
31 – 34	87 – 94	1.08
35 – 38	95 – 100	1.09
39 – 42	101 – 108	1.10
43 – 45	109 – 113	1.11
46 – 48	114 – 119	1.12
49 – 50	120 – 122	1.13

饱和蒸气的介电常数

R410A
温度范围：
-65 – 15 °C / -85 – 59 °F

温度 [℃]	温度 [℉]	制冷剂气体的介电常数 参数 2.5.3 GAS EPS.R
-65 – -47	-85 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -13	-1 – 9	1.05
-12 – -8	10 – 18	1.06
-7 – -4	19 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 4	33 – 40	1.09
5 – 7	41 – 45	1.10
8 – 10	46 – 50	1.11
11 – 12	51 – 54	1.12
13 – 15	55 – 59	1.13

R404A
温度范围：
-60 – 15 °C / -76 – 59 °F

温度 [℃]	温度 [℉]	制冷剂气体的介电常数 参数 2.5.3 GAS EPS.R
-60 – -47	-76 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -14	-1 – 7	1.05
-13 – -9	8 – 16	1.06
-8 – -4	17 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 3	33 – 38	1.09
4 – 6	39 – 43	1.10
7 – 9	44 – 49	1.11
10 – 12	50 – 54	1.12
13 – 15	55 – 59	1.13

R507
温度范围：
-60 – 15 °C / -76 – 59 °F

温度 [℃]	温度 [℉]	制冷剂气体的介电常数 参数 2.5.3 GAS EPS.R
-60 – -48	-76 – -54	1.01
-47 – -36	-53 – -32	1.02
-35 – -28	-31 – -18	1.03
-27 – -21	-17 – -6	1.04
-20 – -15	-17 – -5	1.05
-14 – -10	-4 – 14	1.06
-9 – -6	13 – 22	1.07
-5 – -2	23 – 29	1.08
-1 – 2	30 – 36	1.09
3 – 5	37 – 41	1.10
6 – 8	42 – 47	1.11
9 – 11	48 – 52	1.12
12 – 13	53 – 56	1.13
14 – 15	57 – 59	1.14

语言设置（默认语言：英语）

默认屏幕

AKS 4100
距离
5000 mm

- 按下 ⊙

AKS 4100
1.0.0
快速设置

- 按下 ⊙

AKS 4100
2.0.0
操作员

- 按下 ⊙

AKS 4100
2.0.0
____

输入密码：

⊙ ⊙ ⊙ ⊙ ⊙ ⊙

AKS 4100
2.1.0
信息

- 按下 ⊙ 6次

AKS 4100
2.7.0
显示

- 按下 ⊙

AKS 4100
2.7.1
语言

- 按下 ⊙

AKS 4100
语言
中文

- 按下 ⊙ 或 ⊙ 查看语言选项
  按下 ⊙ 以示确认。

AKS 4100
2.7.1
语言

- 连按 ⊙ 3次

AKS 4100
1.0.0
保存设置 NO

- 按下 ⊙ 或 ⊙ ，在 STORE NO（保存 否）和STORE YES（保存 是）之间进行选择。
  按下 ⊙ 选择 STORE YES（保存 是）

显示默认屏幕：

AKS 4100
距离
5000 mm

语言设置完成

恢复出厂设置

- 进入操作员（查看）菜单（参见第7页）
- 进入参数2.9.4 恢复出厂设置
- 选择恢复出厂设置 是（恢复出厂设置：是）
- 按下 ⊙ 三次，返回默认屏幕

恢复出厂完成

日本語

AKS 4100/4100U は必ず直立管 (柱/バイパス/静止筒) に設置するようにしてください。直立管は以下の場合に使用されます:

- AKS 4100用の整備
- タンク内にアワが多くある場合
- 液体が非常に激しく乱れている場合

冷媒

AKS 4100/4100U は、特に R717 (アンモニア)、HCFC、HFCといった最も広く使用されている冷媒や非腐食性ガス/液体のレベルを計測するために設計されています。

AKS 4100/4100Uは、同軸バージョンではR744(CO_2)にも使用できます。詳細については技術カタログをご覧ください。

基本データ

AKS 4100/4100U は、ループ給電のパッシブ 2 線 4～20 mA センサーです。

供給電圧
14～30 V DC 最大出力 22 mA
負荷
RL [Ω] ≤ ((Uext -14 V)/20 mA).
– デフォルト (エラー出力は 3.6 mA に設定)
RL [Ω] ≤ ((Uext -14 V)/22 mA).
– (エラー出力は 22 mA に設定)

ケーブルグランド
AKS 4100　　PG 13、M20×1.5 ;
　　　　　　(ケーブル径:
　　　　　　6～8 mm (0.24～0.31インチ)
AKS 4100U　½ インチ NPT

端子 (バネ懸架式)
0.5～1.5 mm² (～20-15 AWG)

保護構造
IP 67 (～NEMA タイプ 4X)

冷媒温度
-60 – 100 °C / -76 – 212 °F

冷媒
リストにある冷媒は Danfoss が認定したものです:

R717 / NH_3:	-40 – 50 °C / -40 – 122 °F
R744 / CO_2:	-50 – 15 °C / -58 – 59 °F
HCFC:	
R22:	-50 – 48 °C / -58 – 118 °F
HFC:	
R404A:	-50 – 15 °C / -58 – 59 °F
R410A:	-50 – 15 °C / -58 – 59 °F
R134A	-40 – 50 °C / -40 – 122 °F

(詳細は技術カタログを参照)

周囲温度
-40 – 80 °C / -40 – 175 °F
HMI の場合:-20 – 60 °C / -4 – 140 °F

処理圧力
-1 – 100 barg / -14.5 – 1450 psig

5 m (197 インチ) Ø2 mm (0.08 インチ) ステンレスケーブルによる機械プロセス接続部:
AKS 4100　　G1 インチ管ネジ。
　　　　　　アルミニウム製ガスケット付属
AKS 4100U　¾ インチ NPT
(詳細は技術カタログを参照)

機械設置

機械設置前の準備
信号変換器を機械プロセス接続部から外します (5 mm 六角レンチを使用。図 6 を参照)。赤色の保護カバーを機械プロセス接続部に取り付けて湿気やほこりから保護します。

付属品の内容 (図 1)
① 信号変換器 (HMI がある場合とない場合があります)
② 5 m (197 インチ) Ø2 mm (0.08 インチ) ステンレス ケーブルによる機械プロセス接続部
③ 釣り合いおもり
④ アクセサリ バッグ:
　3 mm ネジ (釣り合いおもり取付用)
　信号変換器を取り付ける前に機械プロセス接続部②を保護するための赤色のカバー
　設定ラベル

注:
直立管の設計ガイドライン:

側面接続管は、直立管内部に出っ張ってはなりません (図 2a)。

側面接続管の推奨径:直立管径 x 0.5 (たとえば直立管の径が DN100 の場合には、側面接続管の径は DN50 以下でなければなりません) (図 **2a**)。

側面接続管についての上記の設計ガイドラインが守 られない場合には、以下のいずれかの方法を推奨します。

1. ディテクションディレーを長くする。パラメータ 2.3.6 ディテクションディレー (パラメータ 2.3.6) を標準の 0 mm から側面上部の接続部の最下点プラス 50 mm まで延ばすことを推奨します (図 2a)。

ディテクションディレー (パラメータ 2.3.6) を変更しても、AKS 4100/4100U の (0%) 4mA および (100%) 20mA 設定を変更する必要はありません。ディテクションディレー内では、計測は 行われません。

2. ケーブルを同軸バージョンに交換する。

直立管は、その全長にわたって同じ直径でなければなりません。直立管の直径が変化する場合には (**図 2b**)、ケーブル バージョンはお勧めできません。同軸バージョンを使用してください。

釣り合いおもり羽根の調整
ガイド付き羽根と管の内壁との間に 5 mmの空間が空くようにします(図 2c 参照)。サイド カッターを使用してガイド付き羽根の形を整え、実際の直立管の直径 に合わせます (図 3 を参照)。
ケーブル プローブの調整

ステンレス ケーブルが折れたりよじれたりしないように注意してください。

すべての計測の開始点を決定する際には、必ず機械プロセス接続部の基準点 (図 4 を参照) を参照してください:
- ケーブルの切断箇所。
- プローブ長 **(図 5 を参照)**
- **スケール** 4 mA **(図 5 を参照)**
- **スケール** 20 mA **(図 5 を参照)**

AKS 4100/4100U の HMI (ヒューマン マシン インターフェイス) をプログラミングする際には、プローブ長、スケール 4 mA、スケール 20 mA を使用します。

図 4 および 5 を参照し、以下の説明に従ってください:
1. 直立管の内側の長さを測定します。
2. ケーブル切断前の準備
   既知のデータ:
   釣り合いおもり下の空間:20 mm (0.8 インチ)
   釣り合いおもり内の鋼線挿入長:12 mm (0.5 インチ)
   釣り合いおもりの高さ:33 mm (1.3 インチ)

   最大プローブ長 =
   　直立管の内部長さ
   – 釣り合いおもり下の空間 (20 mm (0.8 インチ))

   ケーブル長 =
   　最大プローブ長
   + 釣り合いおもり内の鋼線挿入長さ
   　長 (12 mm (0.5 インチ))
   – 釣り合いおもりの高さ (33 mm (1.3 インチ))
3. ケーブルの切断ポイントを測定します。
   基準ポイント **(図 4) から計測してケーブルを切断します**。
4. 釣り合いおもりをケーブルに取り付け、2本の取り付けネジと3mm六角レンチを使用して固定します **(図 3)**。
5. ネジ切りされた穴を通して釣り合いおもりを下げます。**釣り合いおもりが抵抗なく管を通って下がっていき、ケーブルが直立管または流入管の内壁に触れることなく真っ直ぐになっていることを確認します (図 2a を参照)**。
6. トルクレンチを使用して、機械プロセス接続部**(図1 の項目 2)** を 120 Nm (89 lb/ft) まで締め付けます。

計測範囲の計算

4 mA 設定の最大計測範囲:
　= 最大プローブ長
　– 釣り合いおもりの高さ (33 mm (1.3 インチ))
　– 下部デッドゾーン (**図 5 を参照**)

20 mA 設定の最大計測範囲:
　= 上部デッドゾーン (**図 5 を参照**)

例
既知のデータ:
釣り合いおもり下の空間:20 mm (0.8 インチ)
釣り合いおもり内の鋼線挿入長:12 mm (0.5 インチ)
釣り合いおもりの高さ:33 mm (1.3 インチ)
条件:
工場出荷時の設定を使用
冷媒 = アンモニア
直立管の内部長 = 3100 mm (122 インチ)
最大プローブ長 = 3100 mm – 20 mm = 3080 mm (122 インチ – 0.8 インチ = 121.3 インチ)

ケーブル長:
　最大プローブ長 =
+ 釣り合いおもりの鋼線挿入長 (12 mm (0.5 インチ))
– 釣り合いおもりの高さ (33 mm (1.3 インチ))
　3080 mm + 12 mm – 33 mm = **3059 mm**
　(121.3 インチ + 0.5 インチ – 1.3 インチ = **120.4 インチ)**

4 mA 設定の最大計測範囲:
　最大プローブ長 (3080 mm (121.3 インチ))
– 釣り合いおもりの高さ (33 mm (1.3 インチ))
– 下部デッドゾーン (図 5 を参照)
　(210 mm (8.3 インチ)) = **2837 mm (111.7 インチ)**

20 mA 設定の最大計測範囲:
　= 上部デッドゾーン (図 5 を参照) = 120 mm (4.7 インチ)

AKS 4100/4100U コンバータの取り付け方法 (図 6 を参照)
1. 信号変換器の位置決めネジと通気ネジを5 mm の六角レンチで外してください。
2. 信号変換器を下方向に止まるまで押し込み、機械プロセス接続部と確実に勘合させます。
3. 信号変換器を希望する向きに回します。
4. 位置決めネジを5mmの六角レンチで締め付けてください。
5. 通気ネジを5mmの六角レンチで締め付けてください。

電気接続

出力端子 (図 7 および 8)
1. 電流出力 –
2. 電流出力 +
3. 接地端子

電気接続の手順
1. 2.5 mm 六角レンチを使用してカバー止めを緩めます。
2. 筐体から端子部品カバーを取り外します。
3. ワイヤは端子部品カバーから外してはなりません。
   筐体の横に端子部品カバーを置きます。
4. ワイヤを装置に接続します。
   ケーブル入口金物を締め付けます。
5. 端子部品カバーを筐体に取り付けます。
6. 2.5 mm 六角レンチを使用してカバー止めを緩めます。

起動:
- 変換器を電源に接続します。
- 変換器に通電します。

HMI ディスプレイオプションのある装置のみ:10 秒後に「Starting up」と画面に表示されます。20 秒後にソフトウェアのバージョン番号が画面に表示されます。**30 秒後にデフォルト画面 (図 12) が表示されます。**

変更時の注意点
AKS 41/41U から AKS 4100/4100U

注:
AKS 41/41U は交流と直流の両方の電源に対応していますが、AKS 4100/4100U は直流電源のみで使用します。**図 9 の説明に従ってください。**

コントローラまたはPLCへの接続 図 **10 または 11 の**説明に従ってください。

AKS 4100/4100 がマーカー 1、2、3 のようなエラーを検知した場合には、電流出力は 3.6 mA に設定されます (4 ページ参照)。

クイック セットアップ →

注:
信号変換器は、機械プロセスコネクタ付き、またはなしでプログラムすることができます。

クイック セットアップ (以下の値はすべて一例です)

- 装置を電源に接続します (「電気接続」のセクションを参照してください)。
- を 3 回押します。

AKS 4100
クイックセットアップ?
はい　いいえ

- を押します。

AKS 4100
プローブタイプ
ケーブル式

- または を押して、「シングルケーブル式」、「同軸式 D14」、または「同軸式 D22」を選択します
「**ケーブル式**」を選択し、を押して確定します。

AKS 4100
プローブ長さ
05000 mm

- を押して「プローブ長さ」を変更します
を押してカーソルの位置を変更します。
を押して値を減らすか、を押して値を増やします。
を押して確定します。

AKS 4100
4mA位置
04946 mm

- を押して「4mA位置」に変更します。
を押してカーソルの位置を変更します。
を押して値を減らすか、を押して値を増やします。
を押して確定します。

AKS 4100
20mA位置
00070 mm

- を押して「20mA位置」に変更します。
を押してカーソルの位置を変更します。
を押して値を減らすか、を押して値を増やします。
を押して確定します。

AKS 4100
クイックセットアップ
コンプリート8

- 高速設置が完了するまで 8 秒間待ちます。

AKS 4100
1.0.0
クイックセットアップ?

- を押して確定します。

AKS 4100
1.0.0
保存中止

- またはを押して「保存中止」または「保存」のいずれかを選択します。
を押して確定します。

デフォルト画面が表示されます:

AKS 4100
距離
5000 mm

クイックセットアップ完了

を2回押して設定を確認することができます。

AKS 4100	
ケーブル式	5000 mm
(0%) 4 mA	4877 mm
(100%) 20 mA	120 mm

を押してデフォルト画面に戻ります。

強制的に mA 出力にするには (以下の値はすべて一例です)

デフォルト画面

AKS 4100
距離
5000 mm

- を押します。

AKS 4100
1.0.0
クイックセットアップ

- を押します。

AKS 4100
2.0.0
スーパーバイザー

- を押します。

AKS 4100
2.0.0
____

パスワードを入力します:

AKS 4100
2.1.0
インフォメーション

- を押します。

AKS 4100
2.2.0
テスト

- を押します。

AKS 4100
2.2.1
出力設定

- を押します。

AKS 4100
出力設定
3.5 mA

- を押して値を減らすか、を押して値を増やします。
を押して確定します。

AKS 4100
出力設定
8 mA

- を4回押してデフォルト画面に戻ります。

デフォルト画面が表示されます:

AKS 4100
距離
5000 mm

オプションの手順
直立管の温度条件を理解している場合には、定数 (冷媒ガスの誘電率) を入力 (パラメータ 2.5.3 ガス誘電率) して、上部および下部デッドゾーンの下限値を得ることができます (図 5 を参照)。

冷媒ガスの誘電率の入力方法 (以下の値はすべて一例です)

デフォルト画面

AKS 4100
距離
5000 mm

- を押します。

AKS 4100
1.0.0
クイックセットアップ

- を押します。

AKS 4100
2.0.0
スーパーバイザー

- を押します。

AKS 4100
2.0.0
____

パスワードを入力します:

AKS 4100
2.1.0
インフォメーション

- を 4 回押します。

AKS 4100
2.5.0
アプリケーション

- を押します。

AKS 4100
2.5.1
追従速度

- を 2 回押します。

AKS 4100
2.5.3
ガス誘電率

- を押して「ガス誘電率」を確認/変更します。
(以下の表および 8 ページの表から正しい 値を選択します)
を押してカーソルの位置を変更します。
を押して値を減らすか、を押して値を増やします。

AKS 4100
ガス誘電率
1.066

- を押して確定します。

AKS 4100
2.5.3
ガス誘電率

- を 3 回押します。

AKS 4100
1.0.0
保存中止

- または を押して「保存中止」または「 保存」を選択します。
を押して「保存」を選択します。

デフォルト画面が表示されます:

AKS 4100
距離
5000 mm

冷媒ガスの誘電率の入力完了

飽和蒸気誘電率 (デフォルト値:1.066)

R717 (NH_3)
温度範囲:
-60 – 50 °C / -76 – 122 °F

温度 [°C]	温度 [°F]	誘電率 冷媒ガスの パラメータ 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1.00
-41 – -18	42 – 0	1.01
-17 – -5	1 – 23	1.02
-4 – 4	24 – 39	1.03
5 – 12	40 – 54	1.04
13 – 18	55 – 64	1.05
19 – 24	65 – 75	1.06
25 – 28	76 – 82	1.07
29 – 33	83 – 91	1.08
34 – 37	92 – 99	1.09
38 – 40	100 – 104	1.10
41 – 44	105 – 111	1.11
45 – 47	112 – 117	1.12
48 – 50	118 – 122	1.13

R22
温度範囲:
-60 – 48 °C / -76 – 118 °F

温度 [°C]	温度 [°F]	誘電率 冷媒ガスの パラメータ 2.5.3 GAS EPS.R
-60 – -50	-76 – -58	1.00
-49 – -25	57 – -13	1.01
-24 – -10	-12 – 14	1.02
-9 – 0	15 – 32	1.03
1 – 8	33 – 46	1.04
9 – 15	47 – 59	1.05
16 – 21	60 – 70	1.06
22 – 26	71 – 79	1.07
27 – 31	80 – 88	1.08
32 – 35	89 – 95	1.09
36 – 39	96 – 102	1.10
40 – 42	103 – 108	1.11
43 – 45	109 – 113	1.12
46 – 48	114 – 118	1.13

R744 (CO_2)
温度範囲:
-56 – 15 °C / -69 – 59 °F

温度 [°C]	温度 [°F]	誘電率 冷媒ガスの パラメータ 2.5.3 GAS EPS.R
-56.0 – -42.0	-69 – -43	1.01
-41.0 – -28.0	-42 – -18	1.02
-27.0 – -17.0	-17 – 2	1.03
-16.0 – -9.0	3 – 16	1.04
-8.0 – -3.0	17 – 27	1.05
-2.0 – 2	28 – 36	1.06
3 – 7	37 – 45	1.07
8 – 11	46 – 52	1.08
12 – 14	53 – 58	1.09
15	59	1.10

R134a
温度範囲:
-60 – 50 °C / -76 – 122 °F

温度 [°C]	温度 [°F]	誘電率 冷媒ガスの パラメータ 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1.00
-41 – -18	-42 – -0	1.01
-17 – -4	1 – 25	1.02
-3 – 5	26 – 41	1.03
6 – 13	42 – 56	1.04
14 – 20	57 – 68	1.05
21 – 25	69 – 77	1.06
26 – 30	78 – 86	1.07
31 – 34	87 – 94	1.08
35 – 38	95 – 100	1.09
39 – 42	101 – 108	1.10
43 – 45	109 – 113	1.11
46 – 48	114 – 119	1.12
49 – 50	120 – 122	1.13

飽和蒸気誘電率

R410A

温度範囲:
-65 – 15 °C / -85 – 59 °F

温度 [°C]	温度 [°F]	誘電率 冷媒ガスの パラメータ 2.5.3 GAS EPS.R
-65 – -47	-85 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -13	-1 – 9	1.05
-12 – -8	10 – 18	1.06
-7 – -4	19 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 4	33 – 40	1.09
5 – 7	41 – 45	1.10
8 – 10	46 – 50	1.11
11 – 12	51 – 54	1.12
13 – 15	55 – 59	1.13

R507

温度範囲:
-60 – 15 °C / -76 – 59 °F

温度 [°C]	温度 [°F]	誘電率 冷媒ガスの パラメータ 2.5.3 GAS EPS.R
-60 – -48	-76 – -54	1.01
-47 – -36	-53 – -32	1.02
-35 – -28	-31 – -18	1.03
-27 – -21	-17 – -6	1.04
-20 – -15	-17 – -5	1.05
-14 – -10	-4 – 14	1.06
-9 – -6	13 – 22	1.07
-5 – -2	23 – 29	1.08
-1 – 2	30 – 36	1.09
3 – 5	37 – 41	1.10
6 – 8	42 – 47	1.11
9 – 11	48 – 52	1.12
12 – 13	53 – 56	1.13
14 – 15	57 – 59	1.14

R404A

温度範囲:
-60 – 15 °C / -76 – 59 °F

温度 [°C]	温度 [°F]	誘電率 冷媒ガスの パラメータ 2.5.3 GAS EPS.R
-60 – -47	-76 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -14	-1 – 7	1.05
-13 – -9	8 – 16	1.06
-8 – -4	17 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 3	33 – 38	1.09
4 – 6	39 – 43	1.10
7 – 9	44 – 49	1.11
10 – 12	50 – 54	1.12
13 – 15	55 – 59	1.13

言語設定の変更方法 (デフォルト:日本語)

デフォルト画面

AKS 4100
距離
5000 mm

- を押します。

AKS 4100
1.0.0
クイックセットアップ

- を押します。

AKS 4100
2.0.0
スーパーバイザー

- を押します。

AKS 4100
2.0.0
____

パスワードを入力します:

AKS 4100
2.1.0
インフォメーション

- を 6 回押します。

AKS 4100
2.7.0
表示

- を押します。

AKS 4100
2.7.1
言語

- を押します。

AKS 4100
言語
日本語

- または を押して言語の選択肢を表示します。
  を押して確定します。

AKS 4100
2.7.1
言語

- を 3 回押します。

AKS 4100
1.0.0
保存中止

- または を押して「保存中止」または「保存」を選択します。
  を押して「保存実行」を選択します。
  デフォルト画面が表示されます:

AKS 4100
距離
5000 mm

言語設定完了

工場出荷時の設定にリセットする

- 「スーパーバイザー」メニューに移動します (7 ペー ジを参照)。
- パラメータ 2.9.4 の工場設定リセットに移動します。
- 「工場設定リセット」を選択します。
- を 3 回押してデフォルト画面に戻ります。

工場出荷時状態へのリセットが完了します。

РУССКИЙ

Уровнеме AKS 4100/4100U тросовой версии, всегда устанавливается на вертикальной трубе (измерительной колонке и т.д.). Коаксиальная версия уровнемера AKS 4100/4100U допускает установку непосредственно на сасуд.

Хладогенты

AKS 4100/4100U тросовой версии, специально разработан для применения со всеми широко распространёнными хладагентами, включая R717 (аммиак), HCFC, HFC, а также неагрессив-ными газами и жидкостями (кроме CO_2). Для CO_2 необходимо применять коаксиальную версию уровнемера.

Основные тех. характеристики

AKS 4100/4100U это пассивный 2-проводной, уровнемер.

Питающее напряжение
14-30 В постоянного тока (мин./макс. величина) для тока
22 мА на выходе.

Нагрузка
RL [Ω] ≤ ((Uвнеш. -14 В)/20 мА).
– По умолчанию (значение выходного сигнала, при котором выдаётся сообщение об ошибке, установлено на 3,6 мА)
RL [Ω] ≤ ((Uвнеш. -14 В)/22 мА).
– (значение выходного сигнала, при котором выдаётся сообщение об ошибке, установлено на 22 мА)

Кабельный ввод
AKS 4100 PG 13, M20×1.5 ;
(диаметр кабеля: 6-8 мм (0.24-0.31".)
AKS 4100U ½". NPT

Клеммы (с подпружиненными зажимами)
0.5-1.5 мм² (~20-15 AWG)

Степень защиты
IP 67 (~NEMA тип 4X)

Refrigerant temperature
-60 – 100 °C / -76 – 212 °F

Хладагенты
Использование данных датчиков с перечисленными далее хладагентами опробовано и одобрено компанией «Данфосс»:

R717 / NH_3:	-40 – 50 °C / -40 – 122 °F
R744 / CO_2:	-50 – 15 °C / -58 – 59 °F
HCFC:	
R22:	-50 – 48 °C / -58 – 118 °F
HFC:	
R404A:	-50 – 15 °C / -58 – 59 °F
R410A:	-50 – 15 °C / -58 – 59 °F
R134A	-40 – 50 °C / -40 – 122 °F

(Более подробно см. в Техническом описании)

Температура окружающей среды:
-40 – 80 °C / -40 – 175 °F
Для HMI : -20 – 60 °C / -4 – 140 °F

Рабочее давление:
от –1 изб. до 100 изб. (от –14.5 фунт/дюйм² до 1450 фунт/дюйм²)

Механическое соединение тросовой модификации 5 м (197" .) Ø2 (0.08".)

- для датчика AKS 4100: резьбовое соединение с трубной резьбой G 1". В комплект поставки входит алюминиевая прокладка.
- для датчика AKS 4100U: резьбовое соединение с трубной резьбой ¾ " NPT.

Монтаж устройства

Подготовка к монтажу уровнемера
Отсоедините преобразователь сигнала от штуцера (используте ключ на 5 мм, **См. Рис. 6**). Штуцер закройте заглушкой, поставляемой в комплекте. Это позволит избежать загрязне-ния электрических контактов.

Комплектность (Рис. 1)
① Преобразователь сигнала (с или без HMI)
② Штуцер с 5 метровым тросом (5 м (197".) Ø2 мм (0.08 ".)) из нержавеющей стали
③ Центровочный груз
④ Дополнительные принадлежности:
– комплект винтов, размер 3 мм;
– красная крышка, служащая для защиты соединительного штуцера ② до того момента, пока к нему не будет присоединён преобразователь сигналов;
– бирка с данными по настройке датчика.

Внимание:
Рекомендации по конструкции измерительных колонок:
Подсоединительные патрубки **не должны** находиться внутри колонок (**Рис. 2a**)

Для изготовления присоединительных патрубков, рекомендуется применять трубы диаметр которых равен 0,5 x диаметра измерительной колонки, или меньше (т.е для Ду 100 присоединительный патрубок должен быть Ду 50 или меньше) (**Рис. 2a**).

В случае не выполнения, приведенных выше рекомендаций, необходимо выбрать одно из приведенных ниже решений:

1. Увеличить Задержку Обнаружения. Параметр 2.3.6. Мы рекомендуем изменить параметр (по умолчанию 0 мм) на значение равное растоянию до нижнего края верхнего присоединительного патрубка плюс 50 мм. (**Рис. 2a**).

   После увеличения Задержки Обнаружения (параметр 2.3.6), нет необходимости в изме-нении значений 4мА (0%) и 20мА (100%) предварительно настроенных. Так как в данной зоне не будут производиться измерения.
2. Замена тросовой версии на коаксиальную. В измерительная колонка **ДОЛЖНА** иметь один и тот же диаметр по всей своей длинне. Если же диамет различается (**Рис. 2b**), то тросовую версию применять не рекомен-дуется, следует применить коаксиальную версию уровнемера.

Обрезка лопастей центровочного груза
Необходимо оставить зазор, равный 5 мм, между лопастями центровочного груза и внутренней стенкой колонки **(см. Рис. 2c)**. Используя бокорезы обрежде лопасти до необходимого размера **(см. Рис. 3)**.

Подготовка троса к установке

Обратите внимание что трос не должен и меть замятостей и перегибов, по всей своей длине.

Всегда в качестве начальной точки измерений используйте конец резьбы **Штуцера,** как показанно на **Рис. 4**.:
- Обрезка троса.
- Длинна троса **(см. Рис. 5)**
- Точка 4 мА **(см. Рис. 5)**
- Точка 20 мА **(см. Рис. 5)**

Обратите внимание, что длина троса, значе- ния верхней точки 20 мА и нижней 4 мА точки. Необходимы при програмировании уровне-мера по средствам HIM

Следуйте инструкции и **см. Рис. 4 и 5**:
1. Измерьте внутреннюю длинну колонки.
2. *Перед обрезкой троса*
   Дано:
   Растояние от груза до дна колонки: 20 мм (0.8")
   Длина троса зажимаемого в грузе:
   (12 мм (0.5")
   Высота центровочного груза: 33 мм (1.3")

   Максимальная измеряемая длина =
   Внутренняя длина колонки
   – Растояние от груза до дна колонки (20 мм (0.8"))

   Длина троса =
   Максимальная измеряемая длина
   + Длина троса зажимаемого в грузе (12 мм (0.5"))
   – Высота центровочного груза: 33 мм (1.3")
3. Вычислите необходимую длину кабеля. Отмерьте необходимое значение от начальнойточки **(Рис. 4)** и отрежте трос.
4. Установите центровочный груз на тросе, закрепите при помощи 2х винтов на 3 мм.
5. Пропустите груз через штуцер измерительной колонки. **Убедитесь что груз проходит по калонке без сопротивле-ния и трос полностью натянут (не касается стенок колонки)(см. Рис. 2a).**
6. Используя манаметрический ключ затените штуцер **(Рис. 1, поз.2)** с усилием 120 Нм. **Расчет диапазона измерения**

4 мА *определение значения точки =*
Максимальная измеряемая длина
– Высота центровочного груза: 33 мм (1.3")
– *Нижняя мертвая зона* (см. Рис. 5)

20 мА *определение значения точки =*
Верхняя мертвая зона (см. Рис. 5)

Пример
Дано:
Растояние от груза до дна колонки: 20 мм (0.8")Длина троса зажимаемого в грузе: 12 мм (0.5") Высота центровочного груза: 33 мм (1.3")
Используется заводская настройка
Хладогент = Аммиак
Внутренняя длина колонки = 3100 мм (122")
Максимальная измеряемая длина =
3100 мм – 20 мм = 3080 мм
(122" – 0.8" = 121.3")

Длина троса =
Максимальная измеряемая длина
+ Длина троса зажимаемого в грузе (12 мм (0.5"))
– Высота центровочного груза: 33 мм (1.3")
3080 мм + 12 мм – 33 = **<u>3059 мм</u>**
(121.3" + 0.5" – 1.3" = **<u>120.4")</u>**

4 мА *определение значения точки =*
Максимальная измеряемая длина (3080 мм (121.3"))
– Высота центровочного груза (33 мм (1.3"))
– Нижняя мертвая зона (см. Рис. 5) (210 мм (8.3"))
= **<u>2837 мм (111.7")</u>**

20 мА *определение значения точки =*
Верхняя мертвая точка (см. Рис. 5)
= **<u>120 мм (4.7")</u>**

Установка преобразователя сигнала AKS 4100/4100U (см. Рис. 6)

1. Открутите крережный и вентиляционный винты на преобразователе сигнала при помощи шестигранного ключа 5 мм.
2. Прижмите преобразователь сигнала, чтобы он плотно сидел на штуцере волновода.
3. Поверните преобразователь сигнала в требуемое положение.
4. Затяните крепежный винт шестигранным ключом 5 мм.
5. Затяните вентиляционный винт шестигранным ключом 5 мм.

Электрическое подключения

Выходные клемы (Рис. 7 и 8):
1. Выходной ток –
2. Выходной ток +
3. Клема заземления

Порядок электрического подключения
1. Ослабьте стопорный винт крышки при помощи ключа-шестигранника, имеющего размер 2,5 мм.
2. Снимите крышку от клеммной коробки. Положите крышку клеммной коробки рядом с корпусом.
3. Подключите провода к прибору. Затяните кабельный ввод.
4. Подключите провода к прибору. Затяните кабельный ввод.
5. Установите крышку клеммной коробки на корпус.
6. Затяните стопорный винт при помощи ключа-шестигранника, имеющего размер 2.5 мм

Включение:
- Подсоедините питание.

Только для уровнимеров с HMI: После 10 секунд на дисплее отобразится "Starting up". А после 20 секунд на дисплее отобразится версия прошивки уровнемера. Через 30 секунд дисплей перейдет в рабочий режим **(Рис. 12)**.

При замене AKS 41/41U на AKS 4100/4100U

Внимание:
AKS 41/41U мог работать от двух типов питания AC и DC в свою очередь AKS 4100/4100U использует только тип DC
Пример подключения см. Рис. 9.

Подключение к контройлеру или PLC
Следуйте инструкциям на **Рис. 10 и 11**.

В случае если AKS 4100/4100U во время работы обнаружит ошибку и отобразит 1, 2 или 3 (см. стр. 4 пункт №5). То выходной сигнал будет равен 3.6 мА.

БЫСТРАЯ НАСТРОЙКА →

Внимание:
Преобразователь сигнала может быть настроен как в сборе со штуцером так и без.

Быстрая настройка (все значения приведены в качестве примера)

- Подсоедините питание (см. раздел "Электрическое подсоединение".
- Нажмите ⊙ 3 раза.

```
AKS 4100
БЫСТРАЯ НАСТРОЙКА?
ДА                НЕТ
```

- Нажмите ⊙

```
AKS 4100
ВЫБРАТЬ ТИП
ТРОСОВЫЙ
```

- Нажмите ⊙ или ⊙ для выбора типа тросовый, коак. D14 и коак. D22.
Выберите тросовыйи нажмите ⊙.

```
AKS 4100
Длина сенсора
05000 mm
```

- Нажмите ⊙ для изменения длины троса.
Нажмите ⊙ для изменения положения курсора.
Нажмите ⊙ или ⊙ для изменения значения.
Нажмите для подтверждения ⊙.

```
AKS 4100
Шкала 4мА
04946 mm
```

- Нажмите ⊙ для изменения значения точки 4 mA.
Нажмите ⊙ для изменения положения курсора.
Нажмите ⊙ или ⊙ для изменения значения.
Нажмите для подтверждения ⊙.

```
AKS 4100
Шкала 20 мА
00070 mm
```

- Нажмите ⊙ для изменения значения точки 20 mA.
Нажмите ⊙ для изменения положения курсора.
Нажмите ⊙ или ⊙ для изменения значения.
Нажмите для подтверждения ⊙.

```
AKS 4100
БЫСТРАЯ НАСТРОЙКА
ЗАВЕРШЕНО ЧЕРЕЗ 8
```

- После окончания Быстрой настройки необходимо 8 секунд для сохранения изменений.

```
AKS 4100
1.0.0
БЫСТРАЯ НАСТРОЙКА
```

- Нажмите ⊙ для подтверждения.

```
AKS 4100
1.0.0
Сохранить НЕТ
```

- Нажмите ⊙ или ⊙ для выбора Сохранить НЕТ или Сохранить ДА.
Нажмите для подтверждения ⊙.

Появится экран со значениями по умолчанию:

```
AKS 4100
Дистанция
5000 mm
```

Быстрая настройка закончина

Для проверки настроенных параметров дважды нажмите ⊙.

```
AKS 4100
ТРОСОВЫЙ        5000 mm
(0%) 4 mA       4877 mm
(100%) 20 mA     120 mm
```

Нажмите ⊙ ⊙ ⊙ для возврата в рабочее меню уровнемера.

Проверка выходного сигнала мА (все значения приведены в качевстве примера)

Начальное меню

```
AKS 4100
Дистанция
5000 mm
```

- Нажмите ⊙

```
AKS 4100
1.0.0
БЫСТРАЯ НАСТРОЙКА
```

- Нажмите ⊙

```
AKS 4100
2.0.0
Супервизор
```

- Нажмите ⊙

```
AKS 4100
2.0.0
___________
```

Введите пароль:
⊙ ⊙ ⊙ ⊙ ⊙ ⊙

```
AKS 4100
2.1.0
Информация
```

- Нажмите ⊙

```
AKS 4100
2.2.0
Тестирование
```

- Нажмите ⊙

```
AKS 4100
2.2.1
Тест вых.тока
```

- Нажмите ⊙

```
AKS 4100
Тест вых.тока
3.5 mA
```

- Нажмите ⊙ или ⊙ для изменения значения.
Нажмите для подтверждения ⊙.

```
AKS 4100
Тест вых.тока
8 mA
```

- Нажмите 4 раза ⊙ для возврата в рабочее меню уровнемера.

Начальное меню:

```
AKS 4100
Дистанция
5000 mm
```

Проверка выходного сигнала закончена и отключена.

Дополнительная настройка

Если известны температурные условия в трубе, то может быть введена величина диэлектрической проницаемости хладагента (параметр 2.5.3 GAS EPS.R). Это поможет сократить размеры верхней и нижней мёртвых зон

Введение величины диэлектрической проницаемости (все величины приведены ниже только в качестве примера)

Начальное меню

AKS 4100
Дистанция
5000 mm

- Нажмите ▶

AKS 4100
1.0.0
БЫСТРАЯ НАСТРОЙКА

- Нажмите ▲

AKS 4100
2.0.0
Супервизор

- Нажмите ▶

AKS 4100
2.0.0
____

Введите пароль:
▶ ↵ ▼ ▲ ▶ ↵

AKS 4100
2.1.0
Информация

- Нажмите 4 раза ▲

AKS 4100
2.5.0
Применение

- Нажмите ▶

AKS 4100
2.5.1
Скор.слежения

- Нажмите 2 раза ▲

AKS 4100
2.5.3
Er газа

- Нажмите ▶, чтобы изменить величину диэлектрической проницаемости (Er газа). Выберите соответствующую величину из таблиц на стр. 7 и 8
Нажмите ▶ для изменения положения курсора.
Нажмите ▼ или ▲ для изменения значения.

AKS 4100
Er газа
1.066

- Нажмите для подтверждения ↵.

AKS 4100
2.5.3
Er газа

- Нажмите 3 раза ↵.

AKS 4100
1.0.0
Сохранить НЕТ

- Нажмите ▼ или ▲ для выбора Сохранить НЕТ или Сохранить ДА.
Нажмите для подтверждения ↵.

Появится экран со значениями по умолчанию:

AKS 4100
Дистанция
5000 mm

Введение величины диэлектрической проницаемости закончено.

Диэлектрическая проницаемость насыщенного пара (величина, принятая по умолчанию: 1,006)

R717 (NH_3)

Температурный диапазон:
-60 – 50 °C / -76 – 122 °F

Температура [°C]	Температура [°F]	Диэлектрическая проницаемость. Параметр 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1.00
-41 – -18	42 – 0	1.01
-17 – -5	1 – 23	1.02
-4 – 4	24 – 39	1.03
5 – 12	40 – 54	1.04
13 – 18	55 – 64	1.05
19 – 24	65 – 75	1.06
25 – 28	76 – 82	1.07
29 – 33	83 – 91	1.08
34 – 37	92 – 99	1.09
38 – 40	100 – 104	1.10
41 – 44	105 – 111	1.11
45 – 47	112 – 117	1.12
48 – 50	118 – 122	1.13

R22

Температурный диапазон:
-60 – 48 °C / -76 – 118 °F

Температура [°C]	Температура [°F]	Диэлектрическая проницаемость. Параметр 2.5.3 GAS EPS.R
-60 – -50	-76 – -58	1.00
-49 – -25	57 – -13	1.01
-24 – -10	-12 – 14	1.02
-9 – 0	15 – 32	1.03
1 – 8	33 – 46	1.04
9 – 15	47 – 59	1.05
16 – 21	60 – 70	1.06
22 – 26	71 – 79	1.07
27 – 31	80 – 88	1.08
32 – 35	89 – 95	1.09
36 – 39	96 – 102	1.10
40 – 42	103 – 108	1.11
43 – 45	109 – 113	1.12
46 – 48	114 – 118	1.13

R744 (CO_2)

Температурный диапазон:
-56 – 15 °C / -69 – 59 °F

Температура [°C]	Температура [°F]	Диэлектрическая проницаемость. Параметр 2.5.3 GAS EPS.R
-56.0 – -42.0	-69 – -43	1.01
-41.0 – -28.0	-42 – -18	1.02
-27.0 – -17.0	-17 – 2	1.03
-16.0 – -9.0	3 – 16	1.04
-8.0 – -3.0	17 – 27	1.05
-2.0 – 2	28 – 36	1.06
3 – 7	37 – 45	1.07
8 – 11	46 – 52	1.08
12 – 14	53 – 58	1.09
15	59	1.10

R134a

Температурный диапазон:
-60 – 50 °C / -76 – 122 °F

Температура [°C]	Температура [°F]	Диэлектрическая проницаемость. Параметр 2.5.3 GAS EPS.R
-60 – -42	-76 – -43	1.00
-41 – -18	-42 – -0	1.01
-17 – -4	1 – 25	1.02
-3 – 5	26 – 41	1.03
6 – 13	42 – 56	1.04
14 – 20	57 – 68	1.05
21 – 25	69 – 77	1.06
26 – 30	78 – 86	1.07
31 – 34	87 – 94	1.08
35 – 38	95 – 100	1.09
39 – 42	101 – 108	1.10
43 – 45	109 – 113	1.11
46 – 48	114 – 119	1.12
49 – 50	120 – 122	1.13

Диэлектрическая проницаемость насыщенного пара (величина, принятая по умолчанию: 1,006)

R410A

Температурный диапазон:
-65 – 15 °C / -85 – 59 °F

Температура [°C]	Температура [°F]	Диэлектрическая проницаемость. Параметр 2.5.3 GAS EPS.R
-65 – -47	-85 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -13	-1 – 9	1.05
-12 – -8	10 – 18	1.06
-7 – -4	19 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 4	33 – 40	1.09
5 – 7	41 – 45	1.10
8 – 10	46 – 50	1.11
11 – 12	51 – 54	1.12
13 – 15	55 – 59	1.13

R507

Температурный диапазон:
-60 – 15 °C / -76 – 59 °F

Температура [°C]	Температура [°F]	Dielectric constant of refrigerant gas Parameter 2.5.3 GAS EPS.R
-60 – -48	-76 – -54	1.01
-47 – -36	-53 – -32	1.02
-35 – -28	-31 – -18	1.03
-27 – -21	-17 – -6	1.04
-20 – -15	-17 – -5	1.05
-14 – -10	-4 – 14	1.06
-9 – -6	13 – 22	1.07
-5 – -2	23 – 29	1.08
-1 – 2	30 – 36	1.09
3 – 5	37 – 41	1.10
6 – 8	42 – 47	1.11
9 – 11	48 – 52	1.12
12 – 13	53 – 56	1.13
14 – 15	57 – 59	1.14

R404A

Температурный диапазон:
-60 – 15 °C / -76 – 59 °F

Температура [°C]	Температура [°F]	Диэлектрическая проницаемость. Параметр 2.5.3 GAS EPS.R
-60 – -47	-76 – -52	1.01
-46 – -35	-51 – -31	1.02
-34 – -26	-30 – -14	1.03
-25 – -19	-13 – -2	1.04
-18 – -14	-1 – 7	1.05
-13 – -9	8 – 16	1.06
-8 – -4	17 – 25	1.07
-3 – 0	26 – 32	1.08
1 – 3	33 – 38	1.09
4 – 6	39 – 43	1.10
7 – 9	44 – 49	1.11
10 – 12	50 – 54	1.12
13 – 15	55 – 59	1.13

Как изменить язык вывода информации на экран (по умолчанию задан английский язык)

Начальное меню

AKS 4100
Дистанция
5000 mm

- Нажмите →

AKS 4100
1.0.0
БЫСТРАЯ НАСТРОЙКА

- Нажмите ▲

AKS 4100
2.0.0
Супервизор

- Нажмите →

AKS 4100
2.0.0
____

Enter password:
→ ↵ ▼ ▲ → ↵

AKS 4100
2.1.0
Информация

- Нажмите 6 раз ▲

AKS 4100
2.7.0
Изображение

- Нажмите →

AKS 4100
2.7.1
LANGUAGE

- Нажмите →

AKS 4100
Язык
Русский

- Нажмите ▼ или ▲ для выбора языка
  Нажмите для подтверждения ↵.

AKS 4100
2.7.1
Язык

- Нажмите 3 раза ↵

AKS 4100
1.0.0
Сохранить НЕТ

- Нажмите ▼ или ▲ для выбора
  Сохранить НЕТ или Сохранить ДА.
  Нажмите для подтверждения ↵.

AKS 4100
Дистанция
5000 mm

Изменение языка закончено.

Возврат к заводским настройкам

- Войдите в меню **Супервизор** (см. стр. 7).
- Выберите параметр 2.9.4 завод.настр.
- Выбирите завод.настр. ДА
- Нажмите 3 раза ↵.

Возврат к заводским настройкам закончен.

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Data Sheet

Trouble shooting - Liquid level sensors Type **AKS 4100** and **AKS 4100U**

Designed specifically to measure liquid levels in a wide range of refrigeration applications

The AKS 4100/4100U liquid level sensor is designed specifically to measure liquid levels in a wide range of refrigeration applications.

The AKS 4100/4100U liquid level sensor is based on a proven techhology called Time Domain Reflectometry (TDR) or Guided Micro Wave.

AKS 4100/4100U liquid level sensor can be used to measure the liquid level of many different refrigerants in vessels, accumulators, receivers, standpipes, etc.

The electrical output is a 2-wired, loop powered 4–20 mA output signal, which is proportional to the refrigerant liquid level.

Product specification

Compatibility

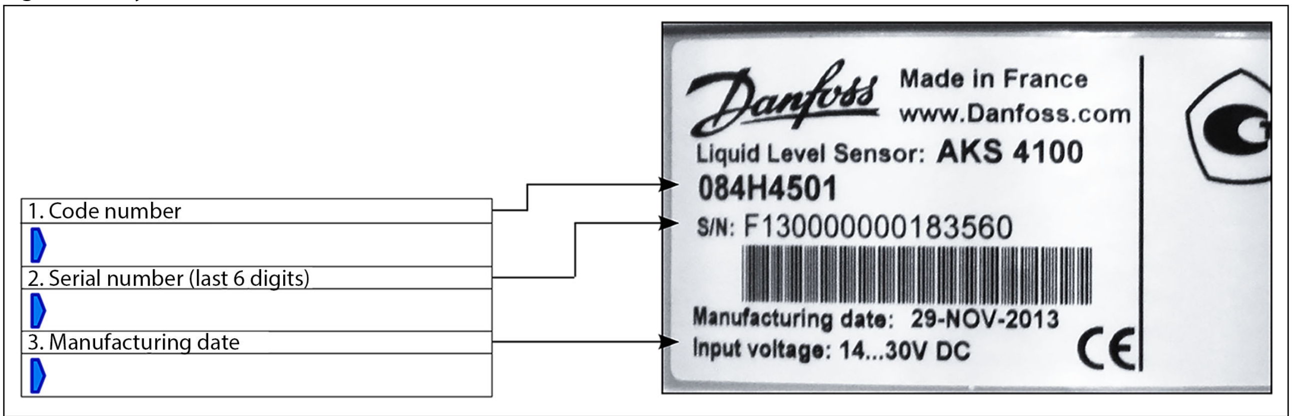
Product identification

Figure 1: Product identification

From the gray label the readings listed below must be noted.

These are used for on-site identification of hardware/software compatibility and for dialog with Danfoss:

Figure 2: Gray label



HMI display readings

Figure 3: Display after power on

The label readings above together with the software stated configuration will identify any possible compatibility problem between the HMI and the converter.

To manually get the software versions of the converter, sensor and HMI please follow the steps on page 2.

Versions

Path to software versions

Please follow the commands to the right to get to the Information menus 2.1.2, 2.1.3 and 2.1.4 and note the versions.

Figure 4: Information menu

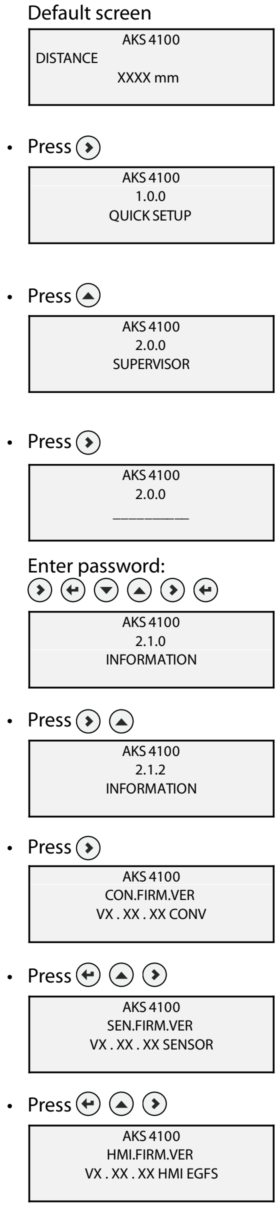


Figure 5: Note the versions

VX . XX . XX	CONV. (Converter)	Menu 2.1.2
VX . XX . XX	SENSOR	Menu 2.1.3
VX . XX . XX	HMI	Menu 2.1.4

Factory software combinations

To the right is the list of the original paired software versions. These combinations are fully functional.

Table 1: Software versions

Manufacturing date Product label	HMI Menu 2.1.4	Converter Menu 2.1.2
31-Oct-11	1.06	1.04
20-Feb-12	1.07	1.06.01
14-May-12	1.08	1.06.01
29-May-12	1.08	1.07.01A
27-Sep-12	1.09	1.07.02
17-Sep-13	1.1	1.08.01

Software backwards compatibility

If, for some reason, the HMI or the converter has been replaced, other version combinations could have been introduced.

Between the different software versions there is to some extend backwards compatibility.

Below table includes green boxes for the allowed version combinations and red boxes for the not allowed combinations.

Table 2: Software versions compatibility

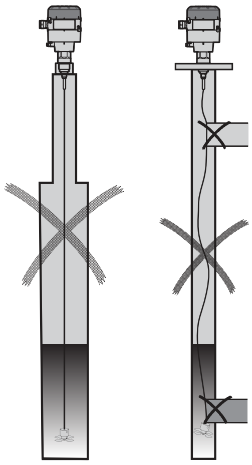
HMI	Converter				
	1.04	1.06.01	1.07.01A	1.07.02	1.08.01
1.06					
1.07					
1.08					
1.09					
1.10.00 EGFS					
1.10.00 ERCJ					

NOTE:
To avoid any problem derived from non compatible software versions it is very important to verify compatibility in the table. In case of any conflict between the two versions Danfoss recommends to replace the HMI display to an allowed software version.

Mechanical

Cable version

Figure 6: Cable version



The cable version is sensitive to variations of the surrounding geometry and materials.

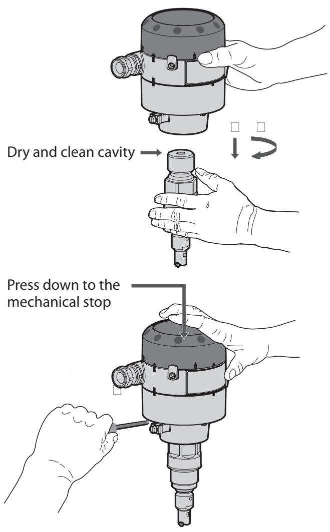
Essential is to avoid these situations:

- Variations in standpipe diameter. In such cases Danfoss recommend the coaxial version.
- Cable not straightening out (missing or hanging counterweight) or cable touching intruding parts/tubing.
- Standpipe made of non-metallic material. Danfoss recommend the coaxial version.

If you are not sure whether the geometry is regular, Danfoss recommend the use of the coaxial version.

Mechanical assembly

Figure 7: Mechanical assembly



It is important to keep the cavity in the upper cable connector dry and clean at all time prior to assembly. For this purpose the packaging includes a red cap to cover the top part.

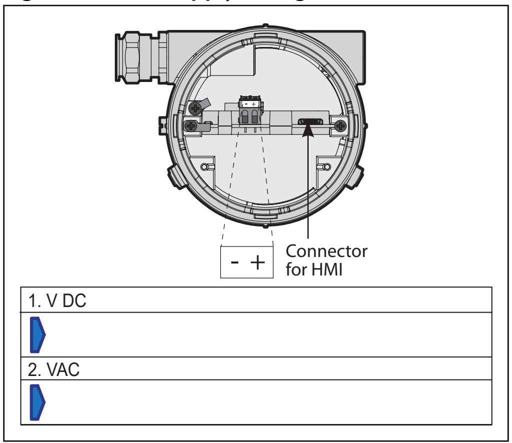
To secure the correct assembly of the converter and cable connector a vertical press to the converter is needed. A mechanical stop indicates the right vertical position.

Remember to tighten the Allen key set bolt with 10 Nm during press down.

Electrical

Electrical requirements (Power supply voltage)

Figure 8: Power supply voltage



1. V DC	Set the Voltmeter to DC Voltage
2. VAC	Switch the Voltmeter to AC Voltage

The Signal converter requires a stable and pure DC power supply.

To check the power quality connect a voltmeter + and – terminals to the + and – terminals of the converter (+ to +, – to –) and keep this wiring throughout below testing.

Set the Voltmeter to DC Voltage. The DC readings must be in the range of 14–30 V.

Switch the Voltmeter to AC Voltage. The AC readings must be lower than 5 V.

If AC readings are above 5 V, the power supply does not meet the required quality.

Signal current

Depending on the wiring between the AKS 4100 and the controller - PLC or EKC/EKE 347 - the signal current can vary.

It is important to have sufficient and identified current to the controller. Below is the path to a controlled forced mA output from the AKS 4100.

Just follow below commands and compare the readings of the forced mA output with the readings in the controller. Remember to measure all current output possibilities of the AKS 4100. These are: 3.5, 4, 6, 8, 10, 12, 14, 16, 18, 20 and 22 mA.

If connected to EKC 347 controller: Read the parameter u30 in EKC for comparison of the forced mA output of the AKS 4100.

If connected to PLC controller: Ask local operator how to read incoming current and compare.

NOTE:
Not all controllers allow a current signal below 4 mA from the AKS 4100. A configured lowest signal of 3.5 mA will in these cases often result in an error reading on the controller.

Figure 9: How to force mA output

How to force mA output

Default screen

AKS 4100
DISTANCE
XXXX mm

- Press (▶)

AKS 4100
1.0.0
QUICK SETUP

- Press (▲)

AKS 4100
2.0.0
SUPERVISOR

- Press (▶)

AKS 4100
2.0.0
____

Enter password:
(▶) (↵) (▼) (▲) (▶) (↵)

AKS 4100
2.1.0
INFORMATION

- Press (▲)

AKS 4100
2.2.0
TESTS

- Press (▶)

AKS 4100
2.2.1
SET OUTPUT

- Press (▶)

AKS 4100
SET OUTPUT
3.5 mA

The HMI reading is now 3.5 mA. Compare with the controller and assure the controller reading is 3.5 mA.

- Press (▼) or (▲) to scroll to any value of mA in the above shown list. Note matching readings from the controller in the table to the right.

AKS 4100
SET OUTPUT
4 mA

⋮

AKS 4100
SET OUTPUT
22 mA

- Press (↵) 4 times to return to default screen.

Default screen appears:

AKS 4100
DISTANCE
XXXX mm

Forced mA completed and disabled

HMI reading mA	Controller reading mA
3.5	
4	
6	
8	
10	
12	
14	
16	
18	
20	
22	

System fault location

Table 3: System fault location

Symptom	Possible cause	Action
Rebooting • Not possible to access the setup menu • Crash or reboot after accessing the setup menu • Showing "Starting up..." continuously • Configurations are not saved	a) Incompatible software versions in HMI and converter respectively. See table on page 2	Identify potential conflict between softwares as described on pages 1 and 2. If not possible to get to Information menu (2.1.2 to 2.1.4) identify the converter from the label and test a HMI which for sure is compatible.
No reaction/ Frozen level Device stays on level ~ 80%, seems "frozen"	a) Non-regular geometry of the standpipe (see page 3) b)Standpipe dimensions out of specification (see Danfoss installation guide) c) Manufacturing date before Sept. 2013	Increase the parameter blocking distance or detection delay up to the level frozen value (see Danfoss installation guide). The value of the blocking distance/ detection delay must be outside the measuring range. If the values of blocking distance/ detection delay cannot be increased further replace the AKS 4100 to a coaxial type (if cable type) or an AKS 4100 (any type) produced after Sep. 2013.
Black-/ interrupted display After HMI connection - no display. - No access to converter setup.	a) Interrupted cable wiring between HMI and converter.	Check the flat cable and connector. Replace the HMI according to compatibility table on page 2.
Unstable/ wrong measurements Periodic fluctuating measurements.	a) Incompatible softwares between HMI and converter. b) Inductor problem on early versions.	Check the compatibility according to table on page 2. Replace with the latest HMI version.
Moisture in display Water leaking through glass sealing.	a) Early versions less water resistant.	Replace with the latest HMI version.
Smell of NH_3 Leakage inside-out of Ammonia Display window discolored/ lost transparency	a) Weak sealings in some early versions.	**HMI[1]**: Replace the HMI with a new version **Mechanical process connection:** *Date code before February 2013:* Replace the mechanical process connection with a new version. *Date code after February 2013:* No need to replace the mechanical process connection

(1) Please note!
- HMI replaced after March 16, 2014 and marked externally with an “A”
- Mechanical process connection replaced after February 2013 and marked with a date code.

A small amount of ammonia may still be smelled when HMI is dismounted. **This represent no safety risk.**
All components are well protected and the AKS 4100/4100U continues to measure and send the 4–20 mA signal corresponding to the liquid level.

Prior to contacting Danfoss; please collect these data:

Collect these data:

Figure 10: Data Form

Data from the grey product label.	**Code number**		
	Serial number		
	Manufacturing date		
Data from HMI menu	**Software versions**	**Converter (2.1.2)**	
		Sensor (2.1.3)	
		HMI (2.1.4)	
	Probe length (2.3.4)		
	Blocking distance (2.3.2)		
	4 mA (2.4.3) =		mm
	20 mA (2.4.4) =		mm
	Coaxial or Cable		
	Refrigerant		
	Refrigerant temperature		
	Refrigerant pressure		

Certificates, declarations, and approvals

Mandatory: Intro to Certificates, declarations, and approvals

The list contains all certificates, declarations, and approvals for this product type. Individual code number may have some or all of these approvals, and certain local approvals may not appear on the list.

Some approvals may change over time. You can check the most current status at danfoss.com or contact your local Danfoss representative if you have any questions.

Valid approvals

Table 4: Valid approvals

File name	Document type	Document topic	Approval authority
GOST FR.C.29.004.A 51938	Measuring - Performance Certificate		
UA.1O146.D.00075-19	UA Declaration	EMCD/LVD	LLC CDC EURO-TYSK
033F0689.AA	EU Declaration	EMC	Danfoss
MD 033F0686.AH	Manufacturers Declaration	PED	Danfoss
033F0695.AA	Manufacturers Declaration	China RoHS	Danfoss
0F18749.513467890YTN	Pressure - Safety Certificate	CRN	TSSA
0F19272.2	Pressure - Safety Certificate	CRN	TSSA

Table 5: Approvals and certification

CE	This device fufills the statutory requirements of the EMC directives. The manufacturer certifies successful testing of the product by applying the CE mark.
	Valid for AKS 4100 - **Not valid for AKS 4100U:** Pattern Approval Certificate of Measuring Instruments for the Russian Federation
EAC	Valid for AKS 4100 - **Not valid for AKS 4100U:** In compliance with EMC regulations in the Russian Federation
EMC	EMC Directives 2004 / 108 / EC and 93 / 68 / EEC in conjunction with EN 61326-1 (2006) and EN 61326-2-3 (2006). The device conforms to these standards if: - the device has a coaxial probe or - the device has a single probe that is installed in a metallic tank
LVD	Low-Voltage Directives 2006 / 95 / EC and 93 / 68 / EEC in conjunction with EN 61010-1 (2001)
NAMUR	NAMUR NE 21 Electromagnetic Compatibility (EMC) of Industrial Process and Laboratory Control Equipment
	NAMUR NE 43 Standardization of the Signal Level for the Failure Information of Digital Transmitters

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